CHAPTER 07

Redox Reactions

In this Chapter...

- Redox Reactions and Oxidation number
- Redox Reactions: Types, Balancing and Applications

Transformation of matter from one form to other takes place through several different types of chemical reactions that involve simultaneous making and breaking of bonds.

Redox reactions is one of the most important category of reactions which plays an important role in our daily life. It is used in several areas like physical, biological, pharmaceutical, industrial, metallurgical, agricultural etc.

Some common examples of redox reactions includes burning of cooking gas, coal, wood, rusting of iron, electroplating, extraction of metals, bleaching and many more.

Recently, environmental issues like hydrogen economy (use of liquid hydrogen as fuel) and development of 'ozone hole' have been considered in redox phenomenon.

Redox Reactions and Oxidation Number

Classical Idea of Redox Reactions : Oxidation and Reduction Reactions

To understand the concept of redox reactions, it is essential to become aware of the oxidation and reduction processes. So, we will first discuss these processes and then redox reactions.

Oxidation

According to classical concept, oxidation is defined as the process that involves.

(i) **Addition of oxygen**

e.g. $2Mg(s) + O_2(g) \longrightarrow 2MgO(s)$

All the combustion processes are examples of it, e.g. combustion of methane to produce CO_2 and water.

$$CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(l)$$

(ii) **Addition of electronegative elements**

e.g. $Mg(s) + \underset{\text{Electronegative element}}{Cl_2(g)} \longrightarrow MgCl_2(s)$

(iii) **Removal of hydrogen**

e.g. $2H_2S(g) + O_2(g) \longrightarrow 2S(s) + 2H_2O(l)$

(iv) **Removal of electropositive elements**

e.g. $2K_4[Fe(CN)_6](aq) + H_2O_2(aq) \longrightarrow 2K_3[Fe(CN)_6](aq) + 2KOH(aq)$

Note From the above discussion, it is clear that oxidation is defined as the process of addition of oxygen/ electronegative element to a substance or removal of hydrogen/electropositive element from a substance.

Reduction

According to classical concept, reduction is defined as a process that involves.

(i) **Addition of hydrogen**

e.g. $\underset{\text{Ethene}}{CH_2{=}CH_2}(g) + H_2(g) \longrightarrow \underset{\text{Ethane}}{CH_3{-}CH_3}(g)$

(ii) **Addition of electropositive element**

e.g. $2HgCl_2(aq) + SnCl_2(aq) \longrightarrow Hg_2Cl_2(s) + SnCl_4(aq)$

(iii) **Removal of oxygen**

e.g. $2HgO(s) \xrightarrow{\Delta} 2Hg(l) + O_2(g)$

(iv) **Removal of electronegative element**

e.g. $2FeCl_3(aq) + H_2(g) \longrightarrow 2FeCl_2(aq) + 2HCl(aq)$

Note The above discussion clearly revealed that reduction is just opposite of oxidation and is defined as the process of removal of **oxygen/electronegative** element from a substance or process of addition of **hydrogen/electropositive element** to a substance.

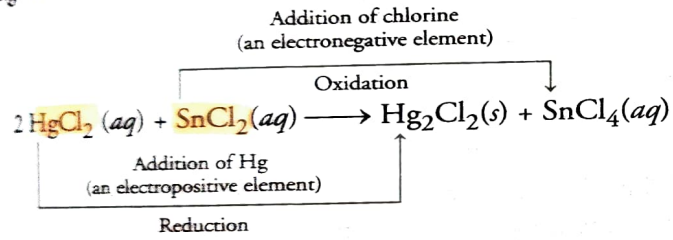
Redox Reactions

The reactions that involve oxidation and reduction as their two half reactions are called redox reactions.

Or

In other words, redox reactions are the reactions in which oxidation and reduction processes occur simultaneously.

e.g. If we consider the following reaction.



Since, in the above reaction oxidation and reduction occur simultaneously, so it is an example of redox reaction.

Note The *'red'* of redox comes from reduction and *'ox'* from oxidation. Always remember that if in a reaction a substance is oxidised, then another gets reduced at the same time or *vice-versa*. Thus, it can be said that oxidation and reduction reactions are complementary, i.e. they always go hand in hand or side by side or oxidation occurs at the cost of reduction.

Oxidising and Reducing Agents

In a redox reaction, the substance which oxidises the other or which itself undergoes reduction is called the oxidising agent. Whereas, the substance that reduces the other and itself undergoes oxidation is called the reducing agent.

Thus, in the given example, $HgCl_2$ is an oxidising agent (as it reduces) and $SnCl_2$ is a reducing agent (as it oxidises).

The Common Oxidising/Reducing Agents and their Reduced/Oxidised Forms

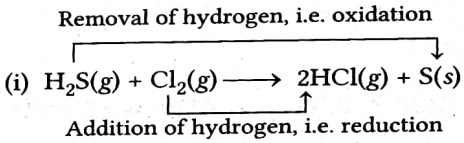
Oxidising agent	Reduced form
HNO_3 (conc.)	$NO_2 + H_2O$
HNO_3 (dil.)	$NO + H_2O$
MnO_4^- (in acidic medium)	Mn^{2+}
MnO_4^- (in basic medium)	MnO_2
$Cr_2O_7^{2-}$ or CrO_4^{2-}	$Cr^{3+} + H_2O$
F_2, Cl_2, Br_2, I_2	F^-, Cl^-, Br^-, I^-
Fe^{3+}	Fe^{2+}
Sn^{4+}	Sn^{2+}
Reducing agent	**Oxidised form**
Metals	Respective cations
H_2S	S or SO_2 or SO_4^{2-}
HCl, HBr, HI	Cl_2, Br_2, I_2
Fe^{2+}	Fe^{3+}
Sn^{2+}	Sn^{4+}
$C_2O_4^{2-}$ (oxalate)	$CO_2 + H_2O$

EXAMPLE 1. In the reactions given below, identify the species undergoing oxidation and reduction.

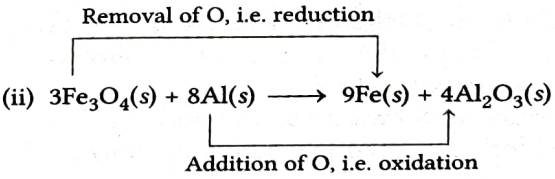
(i) $H_2S(g) + Cl_2(g) \longrightarrow 2HCl(g) + S(s)$

(ii) $3Fe_3O_4(s) + 8\,Al(s) \longrightarrow 9\,Fe(s) + 4Al_2O_3(s)$ **NCERT**

Sol.



Thus, here H_2S undergoes oxidation and Cl_2 undergoes reduction.



Thus, here Fe_3O_4 undergoes reduction and Al undergoes oxidation.

Prediction of oxidation and reduction processes in some reaction is a difficult task on the basis of classical concept so there was a need to find a method with the help of which these can be predicted easily. Such a method was given in terms of electron transfer reactions.

Redox Reactions in Terms of Electron Transfer Reactions

Electronic concept is the modern concept used to define redox reactions. It define oxidation and reduction in terms of transfer of electrons as follows.

Oxidation in Terms of Electron Transfer

According to electronic concept, oxidation is the process which involves loss of electron(s).

The species that undergoes oxidation becomes more positive (i.e. acquires more positive charge) or less negative (i.e. negative charge reduces.).

e.g.

$Na \longrightarrow Na^+ + e^-$ (positive charge increases on Na)

$2Cl^- \longrightarrow Cl_2 + 2e^-$ (negative charge decreases on Cl)

Because of the loss of electrons, the oxidation process can also be named as **de-electronation**.

Reduction in Terms of Electron Transfer

Electronic concept also suggested that reduction is the process that involves gain of electron(s). The species undergoing reduction becomes more negative (i.e. acquires more negative charge) or less positive (i.e. positive charge reduces). e.g.

$Cl_2 + 2e^- \longrightarrow 2Cl^-$ (negative charge increases on Cl)

$Fe^{3+} + e^- \longrightarrow Fe^{2+}$ (positive charge reduces on Fe^{3+})

Because of the gain of electron, reduction process is also called **electronation**.

Redox Reaction in terms of Electron Transfer

The electron lost in an oxidation reaction is simultaneously gained in a reduction reaction. Because of simultaneous loss and gain of electrons in oxidation reduction processes, the **redox reactions**, (or the oxidation-reduction reactions) are also called **electron transfer reactions**.

e.g. The reaction between sodium and chlorine to give sodium chloride is a redox reaction as shown below.

Loss of $2e^-$ (oxidation half reaction)

$$2Na(s) + Cl_2(g) \longrightarrow 2\overset{+}{Na}\,\overset{-}{Cl}\,(s)$$

Gain of $2e^-$ (reduction half reaction)

The two half reactions can be written as

$$2Na(s) \longrightarrow 2Na^+(g) + 2e^-$$

$$Cl_2(g) + 2e^- \longrightarrow 2Cl(g)$$

Similarly, the following two redox reactions can also be illustrated in terms of loss and gain of electrons.

Loss of $2e^-$ (oxidation half reaction)

$$2Na(s) + O_2(g) \longrightarrow \overset{+}{(Na)_2}\,O^{2-}(s)$$

Gain of $2e^-$ (reduction half reaction)

Loss of $2e^-$ (oxidation half reaction)

$$2Na(s) + S(s) \longrightarrow \overset{+}{(Na)_2}\,S^{2-}(s)$$

Gain of $2e^-$ (reduction half reaction)

Oxidising and Reducing Agents in Terms of Electron Transfer

This concept also suggests that a species that gain electrons is called **oxidising agent** or **oxidant** and the one that loss or donate the electrons to the interacting element is called the **reducing agent** or **reductant**.

e.g.

Increase in positive charge, i.e. oxidation or loss of electron

$$Zn(s) + Cu^{2+}(aq) \longrightarrow Cu(s) + Zn^{2+}(aq)$$

Decrease in positive charge, i.e. reduction or gain of electron

Thus, here Zn acts as reducing agent as it reduces Cu^{2+} by providing it two electrons and Cu^{2+} acts as oxidising agent, as it gains (or accepts) two electrons from Zn.

- **Oxidation** Loss of electron(s) by any species.
- **Reduction** Gain of electron(s) by any species.
- **Oxidising agent** Acceptor of electron(s).
- **Reducing agent** Donor of electron(s).

EXAMPLE 2. Justify that the reactions.

(i) $2Na(s) + H_2(g) \longrightarrow 2NaH(s)$ and **NCERT**

(ii) $4Na(s) + O_2(g) \longrightarrow 2Na_2O(s)$ are redox reactions.

Sol. (i) Sodium hydride is an ionic compound, so it can be represented as Na^+H^-. Thus, oxidation half reaction (i.e. increase in positive charge) can be given as

$$2\,Na\,(s) \longrightarrow 2\,Na^+(g) + 2e^- \quad \ldots(i)$$

and the reduction half reaction (i.e. increase in negative charge) is given as

$$H_2(g) + 2e^- \longrightarrow 2H^-(g) \quad \ldots(ii)$$

The sum of above two reactions give the overall reaction. [while adding two reactions, remember that the number of electron lost must be equal to the number of electrons gained. If these two are not

equal, make them equal by multiplying with a suitable integer as done in case of reaction (i).]

Thus, $2\,Na(s) + H_2(g) \longrightarrow 2Na^+(aq) + 2H^-(aq)$

or $2Na(s) + H_2(g) \longrightarrow 2NaH(s)$

Since, this reaction involve oxidation and reduction as its two half reaction, it is a redox reaction.

(ii) Like NaH, sodium oxide is also an ionic compound and have the formula $(Na^+)_2O^{2-}$. Thus, oxidation and reduction half reactions can be written as

$$2\,Na\,(s) \xrightarrow[\text{Loss of } 2e^-]{} 2\,Na^+(g) + 2e^-$$

(Oxidation half reaction) ...(i)

$$O_2(g) + 4\,e^- \xrightarrow[\text{Gain of } 4e^-]{} 2O^{2-}(g)$$

(Reduction half reaction) ...(ii)

To equalise number of electrons, equation (i) is multiplied with 2, and hence, we get

$$4\,Na\,(s) \longrightarrow 4\,Na^+(g) + 4e^- \quad ...(iii)$$

On adding equations (ii) and (iii), we get

$$4\,Na(s) + O_2(g) \longrightarrow 4\,Na^+ + 2\,O^{2-}$$

Or $4Na\,(s) + O_2(g) \longrightarrow 2\,(Na^+)_2O^{2-}(s)$

Since, the above reaction involve oxidation and reduction as its two half reactions, so it is a redox reaction.

Note NaCl and Na_2S being ionic compounds can also be represented as Na^+Cl^- and $(Na^+)_2S^{2-}$.

EXAMPLE 3. Identify the oxidant and reductant in each of the following reactions.

(i) $MnO_4^-(aq) + C_2H_5OH\,(aq) \longrightarrow Mn^{2+}(aq) + CH_3COOH(aq)$

(ii) $Zn\,(s) + 2H^-\,(aq) \longrightarrow Zn^{2+}\,(aq) + H_2\,(g)$

(iii) $PbO\,(s) + C\,(s) \longrightarrow Pb\,(s) + CO\,(g)$

Sol (i) MnO_4^- loses oxygen, therefore it is an oxidant while C_2H_5OH gains oxygen, therefore it is a reductant.

(ii) Zn is reductant while hydrogen (H) is oxidant.

(iii) PbO is oxidant while C is reductant.

EXAMPLE 4. Write the half reactions for the following redox reactions.

(i) $Al(s) + 3Ag^+\,(aq) \longrightarrow Al^{3+}\,(aq) + 3Ag\,(s)$

(ii) $2K\,(s) + Cl_2\,(g) \longrightarrow 2KCl(s)$

Sol (i) $Al\,(s) \longrightarrow Al^{3+}\,(aq) + 3e^-$ (oxidation half reaction)

$3Ag^+\,(aq) \longrightarrow 3Ag(s)$ (reduction half reaction)

(ii) $K \longrightarrow K^+ + e^-$ (oxidation half reaction)

$\frac{1}{2}Cl_2 + e^- \longrightarrow Cl^-$ (reduction half reaction)

Competitive Electron Transfer Reactions

Different elements and ions differ in their electron gaining or losing tendencies.

If these are arranged in decreasing order of electron losing tendency, or increasing order of electron gaining tendency, a series is obtained. This series is known as the **metal activity series** or **electrochemical series.** This series is found helpful to decide the feasibility of a reaction.

The series is

Li> K> Ba > Sr> K> Na> Mg> Al> Mn> Zn>Cr > Fe> Co > Ni > Sb > Pb > H > Cu > Hg > Ag > Pd > Pt > Au

[For details see table at page 311. e.g.

- When a zinc strip is placed into the aqueous solution of copper nitrate, it gets coated with reddish metallic copper and the blue colour of copper nitrate solution disappears.

  The net reaction

$$Zn\,(s) + Cu^{2+}\,(aq) \longrightarrow Zn^{2+}(aq) + Cu\,(s)$$

  (Release of $2e^-$ at Zn; Gain of $2e^-$ at Cu)

  If hydrogen sulphide gas is passed through the colourless solution and the medium is made alkaline by adding ammonia, a white precipitate of zinc sulphide (ZnS) appears, which shows the presence of Zn^{2+} ions in the solution.

  The state of equilibrium for the above reaction can be represented by placing a strip of metallic copper in a zinc sulphate solution.

  No visible reaction is noticed and attempt to detect the presence of Cu^{2+} ions by passing H_2S gas through the solution to produce black colour CuS, does not succeed.

  Thus, the state of equilibrium for the reaction greatly favours the products over the reactants.

- In another example, if copper strip is placed in silver nitrate solution, the strip gets covered with silver and the colour of solution becomes blue.

$$Cu(s) + 2Ag^+\,(aq) \longrightarrow Cu^{2+}(aq) + 2\,Ag(s)$$

(Lose of $2e^-$ at Cu; Gain of $2e^-$ at Ag)

In contrast, when metallic cobalt placed in nickel sulphate solution the following reaction will takes place.

Release of $2e^-$

$$Co(s) + Ni^{2+}(aq) \longrightarrow Co^{2+}(aq) + Ni(s)$$

Gain of $2e^-$

At equilibrium, both $Ni^{2+}(aq)$ and $Co^{2+}(aq)$ are present at moderate concentrations.

The competition for the release of electrons is similar to that of competition for the release of protons among acids. This resemblance suggests that we develop a table of metals and their ions on the basis of their tendency to release electrons in the same way as it is done in the case of acids to indicate the strength of the acids.

These experiments support the activity series (given above). This competition of electrons between the metals help in designing galvanic cells.

Oxidation Number

The concept of electron transfer is found unable to explain the redox changes or electron shift in case of covalent compounds.

To explain these changes a new concept, called oxidation number is introduced. Oxidation number is defined as the charge that an atom of the element has in its ion or appear to have when present in the combined state with other atoms.

Or

In other words, it is also defined as the charge that an atom appear to have in a compound when all other atoms are removed as ions from the compound.

e.g. $\overset{0}{H_2}(s) + \overset{0}{Cl_2}(g) \longrightarrow \overset{+1\ -1}{2HCl}(g)$

$\overset{-4+1}{CH_4} + 4\overset{0}{Cl_2}(g) \longrightarrow \overset{+4\ -1}{CCl_4}(s) + 4\overset{+1\ -1}{HCl}(g)$

Rules for Calculating Oxidation Number

Comparison between the electronegativities of two combining atoms is not an easy task in case of polyatomic molecules. That's why a set of rules has been given to calculate the oxidation number of an element in a given compound or ion.

These rules are given below.

***Rule* I** The oxidation number of an atom in its free or elementary state or in any of its allotropes is zero. e.g. The oxidation of H in H_2, S in S_8, P in P_4, O in O_2 or O_3, C in diamond is zero.

***Rule* II** In case of ions having only one kind of atom, the oxidation number of each atom is equal to charge present on the ion.
e.g. In case of Na^+, Mg^{2+}, Fe^{3+}, Cl^- and O^{2-}, the oxidation states are respectively $+1$, $+2$, $+3$, -1, -2.

***Rule* III** The oxidation number of alkali metals in all of their compounds is always $+1$. Similarly, in case of alkaline earth metals, it is always $+2$. For aluminium, oxidation number is always $+3$.

***Rule* IV** The oxidation number (ON) of oxygen in most of its compounds is -2, with an exception of peroxides and superoxides in which the oxidation number of oxygen is respectively -1 and $-1/2$. e.g. In H_2O_2 (hydrogen peroxide), oxidation number of O is -1 [peroxide linkage is —O—O—] and in KO_2 (potassium superoxide), oxidation number of O is $-1/2$. The second exception appears when oxygen is bonded to fluorine, e.g. oxygen difluoride (OF_2) and dioxygen difluoride (O_2F_2), the oxygen is assigned an oxidation number of +2 and +1, respectively.

***Rule* V** The oxidation number of hydrogen is generally $+1$ with an exception of metallic hydrides like NaH, CaH_2 etc. In these hydrides, oxidation number of hydrogen is -1.

***Rule* VI** The oxidation number of fluorine in all of its compounds is always -1. Other halogens (i.e. chlorine, bromine and iodine) also exhibit -1 oxidation number but it is not always true. In case of oxoacids and oxoanions, halogens (except fluorine) exhibit positive oxidation number.

***Rule* VII** The algebraic sum of the oxidation numbers of all the atoms present in a compound must be equal to zero.

***Rule* VIII** In case of polyatomic ions, the algebraic sum of oxidation number of all the atoms present in the ion must be equal to the charge on the ion. e.g. In case of carbonate ion (CO_3^{2-}) it is equal to -2.

Note Metals generally exhibit positive oxidation states whereas the oxidation state in case of non-metals may be positive or negative. Moreover, transition elements exhibit variable oxidation states.

Highest oxidation number for first two groups = group number.

Highest oxidation number for other groups = group number -10

[Here, group number are in accordance with long form of periodic table]

e.g. In SF_6, highest oxidation number of S = 16 − 10 = 6.

The highest oxidation number exhibited by an element generally increases along a period of the periodic table.

In third period, it increases from 1 to 7.

Group	1	2	13	14	15	16	17
Element	Na	Mg	Al	Si	P	S	Cl
Compound	NaCl	$MgSO_4$	AlF_3	$SiCl_4$	P_4O_{10}	SF_6	$HClO_4$
Highest oxidation number/State of the group element	+1	+2	+3	+4	+5	+6	+7

A term that is often used interchangeably with the oxidation number is the **oxidation state.**

Steps for Calculation of Oxidation Number (ON)

The following steps are involved while calculating the oxidation number of an atom in a given compound/ion.

Step I Write down the formula of given compound/ion leaving some space between the atoms.

Step II Write the oxidation number of each element above its atoms. Write down x above the atom, oxidation number of which we have to find out.

Step III Multiply the oxidation numbers of each element with the number of atoms of that element present in the compound. Enclose the product in a bracket.

Step IV Equate the algebraic sum of the oxidation numbers of all the atoms present in a compound to zero or to the charge in case of ionic species.

Step V Solve the equation obtained for the value of x.

EXAMPLE 5. Calculate the oxidation state of (i) Mn in $KMnO_4$ and (ii) N in NO_3^-.

Sol. (i) Let the oxidation number of Mn in $KMnO_4$ be x.

Writing the oxidation number of each atom above its symbol.

$$\overset{+1}{K}\ \overset{x}{Mn}\ \overset{-2}{O_4}$$

$$[1\times(+1)+(1\times x)+4\times(-2)]=0$$

$$1+x+(-8)=0$$

$$1+x-8=0$$

$$x-7=0$$

$$\therefore\quad x=+7$$

Hence, the oxidation number of Mn in $KMnO_4$ is +7.

(ii) **N in NO_3^-.**

Let the oxidation number of N in NO_3^- be x.

Writing the oxidation number of each atom above its symbol.

$$\overset{x}{N}\ \overset{-2}{O_3^-}$$

$\therefore$ Sum of the oxidation number of all the atoms in

$$NO_3^- = x+3(-2)=x-6$$

But the sum of oxidation numbers of all the atoms in NO_3^- ion is equal to the charge present on it, i.e. −1.

$$\therefore\quad x-6=-1$$

or $$x=+5$$

Thus, the oxidation number of N in NO_3^- is +5.

Paradox of Fractional Oxidation Number

In case of some compounds, oxidation state of an element in the compund is found out to be in fraction.

This fractional oxidation number is indeed the average oxidation state of the element under consideration. It is generally revealed only by the structures that the element in that compound/ion whose fractional oxidation state is obtained is present in more than one oxidation states. This can also be illustrated by the following examples.

(i) **Oxidation state of C in C_3O_2** The oxidation state of C in C_3O_2 = 4 / 3. But as we know it is only the average oxidation state. Thus, the actual oxidation state of each atom is calculated as follows

The structure of C_3O_2 is as

$$O=C=C=C=O$$

The oxidation states of terminal O-atoms is −2, the oxidation states of C bonded with O is +2 and the oxidation state of central C-atom is 0. Thus,

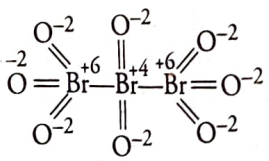
$$\overset{-2}{O}=\overset{+2}{C}=\overset{0}{C}=\overset{+2}{C}=\overset{-2}{O}$$

and the average of oxidation states of 3C- atoms in

$$C_3O_2=\frac{2+2+0}{3}=\frac{4}{3}$$

(ii) **Oxidation state of Br in Br_3O_8 (Tribromooctaoxide)** The average oxidation state of Br (with the help of rules) in Br_3O_8 is 16/3.

The structure of Br_3O_8 is

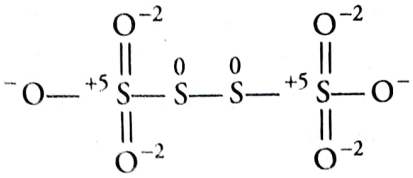


The oxidation state of each O-atom bonding with Br is −2.

Thus, the actual oxidation state of two terminal bromine atoms are +6 and the central bromine atom is +4.

(iii) **Oxidation state of S in $S_4O_6^{2-}$ (Tetrathionate ion)** The average oxidation state of S in $S_4O_6^{2-}$ is 2.5.

The structure of $S_4O_6^{2-}$ ion is



Thus, the actual oxidation state of each terminal S-atom is $+2+2+1=5$ and the central S-atoms is 0.

(iv) **Oxidation number of metals in mixed oxides** Fe_3O_4, Mn_3O_4, Pb_3O_4 etc., all are mixed oxides with actual composition ($FeO \cdot Fe_2O_3$), ($2MnO \cdot MnO_2$), ($2PbO, PbO_2$) respectively.

In FeO, oxidation state of Fe is +2 and in Fe_2O_3, its oxidation state is +3. The average of these two oxidation states is $\frac{5}{3}$ which is also obtained from the rule. Similarly, in MnO, oxidation state of Mn is +2 whereas in MnO_2 it is +4.

Their average in Mn_3O_4 is $\frac{2+2+4}{3}=\frac{8}{3}$.

In red lead (Pb_3O_4), oxidation state of Pb in PbO is +2 and in PbO_2 is +4.

Hence, their average value is $\frac{2+2+1(+4)}{3}=\frac{8}{3}$.

(v) **Oxidation number of Cl in bleaching powder, $CaOCl_2$** The actual composition of $CaOCl_2$ is $Ca^{2+}(OCl^-)Cl^-$, thus the oxidation state of Cl in Cl^- is −1 and of Cl in OCl^- is +1.

Note The oxidation state of O_2^+ and O_2^- are + 1/2 and − 1/2 respectively, i.e. fractional.

Applications of Oxidation Number

The important applications of oxidation numbers are as follows

(i) Stock Notation

This system of nomenclature was introduced by **Alfred Stock**, a German chemist and is called the **Stock notations**.

According to this notation, the oxidation number of an element can be expressed as roman numeral in parenthesis after the symbol of that element in the molecular formula.

e.g. In AuCl (aurous chloride), the oxidation state of Au is +1 and in $AuCl_3$ (auric chloride), the oxidation state of Au is +3. Thus, they can be distinguished as Au (I) Cl and Au (III) Cl or gold (I) chloride and gold (III) chloride.

With the help of Stock notations, it is possible to predict that whether the species is present in oxidised form or reduced form.

Stock Notation of Some Important Compounds

Compound/Formula	Oxidation state of metal	Stock notation
$SnCl_2$ (Stannous chloride)	+ 2	$Sn(II)Cl_2$
$SnCl_4$ (Stannic chloride)	+ 4	$Sn(IV)Cl_4$
$FeSO_4$ (Ferrous sulphate)	+ 2	$Fe(II)SO_4$
V_2O_5 (Vanadium pentoxide)	+ 5	$V_2(V)O_5$
Cr_2O_3 (Chromium trioxide)	+ 3	$Cr_2(III)O_3$

Note PCl_3 and PCl_5 can never be represented by Stock notations because non-metallic elements shows variable valencies.

EXAMPLE 6. Using Stock notation, represent the following compounds $HAuCl_4$, Tl_2O, FeO, Fe_2O_3, CuI, CuO, MnO and MnO_2. **NCERT**

Sol. (i) **$HAuCl_4$**

Let the oxidation state of Au is x.

$$\overset{+1}{H}\quad \overset{x}{Au}\quad \overset{-1}{Cl_4}$$
$$(+1)\quad x\quad 4(-1)$$

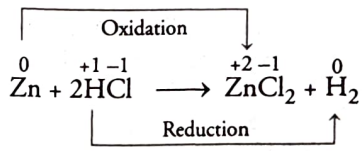
$$(+1)+x+4(-1)=0 \text{ or } x-3=0 \text{ or } x=+3$$

Thus, the Stock notation for the compound is $HAu(III)Cl_4$. Find the Stock notations of other compounds in the same manner as shown above. You will get the answer as $Tl_2(II)O$, $Fe_2(II)O$, $Fe_2(III)O_3$, Cu (I)I, Cu(II)O, Mn(II)O, $Mn(IV)O_2$.

(ii) Prediction of Oxidation-Reduction Reactions

In term of the concept of oxidation number, **oxidation is defined as the process in which the oxidation number of an element increases whereas reduction is the processs in which the oxidation number of an element decreases.**

e.g. Consider the following reaction,



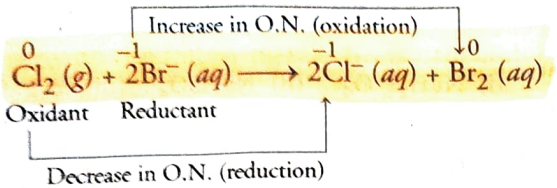
In the above reaction, oxidation number of Zn increases from 0 to +2 but of H decreases from +1 to 0. Thus, here Zn is oxidised whereas HCl is reduced. Hence, it is a redox reaction.

Thus, it is also concluded that the reactions in which the oxidation states of interacting species change simultaneously are called the **redox reactions.**

(iii) Prediction of Oxidising and Reducing Agents

In term of concept of oxidation number (ON), an **oxidising agent** or **oxidant** is the substance, oxidation number of which decreases in the chemical reaction.

It is the substance that itself undergoes reduction whereas **reducing agent** or **reductant** is the substance, oxidation number of which increases in the chemical reaction. Thus, it is the substance that itself undergoes oxidation.
e.g. Consider the following example,



Here oxidation number of Br increases from -1 to 0, so it behaves as a **reductant** whereas the oxidation number of Cl decreases from 0 to -1, so it behaves as an **oxidant.**

Quick Trick

- A substance behaves as an oxidant only when any one of its elements is present in its highest oxidation state. e.g. $KMnO_4$.
- A substance behaves as a reductant only, when any one of its element is present in its lowest possible oxidation state. e.g. H_2S.
- Substance having an element in its intermediate oxidation state can behave as oxidant as well as reductant. e.g. H_2O_2.

EXAMPLE 7. Justify that the reaction,

$$2Cu_2O\,(s) + Cu_2S\,(s) \longrightarrow 6\,Cu\,(s) + SO_2(g)$$

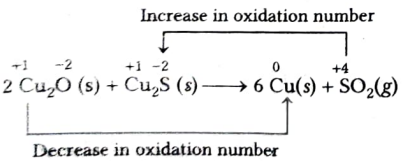
is a redox reaction. Identify the species oxidised/reduced, which acts as an oxidant and which acts as a reductant. **NCERT**

Sol.

Step I Write the oxidation number of each element above it in the given reaction.

e.g. $\overset{+1\ -2}{2Cu_2O}\,(s) + \overset{+1\ -2}{Cu_2S}\,(s) \longrightarrow \overset{0}{6Cu}(s) + \overset{+4\ -2}{SO_2}(g)$

Step II Mark arrow to show the increase and decrease in oxidation number. Also mention it over the arrow.



Step III If the reaction involves increase as well as decrease in the oxidation number of certain elements, then it is called redox reaction, otherwise not.

Since, in the above reaction oxidation state decreases as well as increases, so it is a redox reaction (proved).

Step IV Increase in oxidation number shows oxidation whereas decrease in oxidation number shows reduction.

Thus, predict the species undergoing oxidation and reduction.

Here, oxidation number of S increases from -2 to $+4$, so Cu_2S is oxidised here. Further, oxidation number of Cu decreases for $+1$ to 0, so Cu_2O is reduced.

Thus, here Cu_2S is oxidised and Cu_2O is reduced .

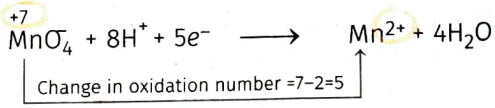
Step V Further predict oxidant and reductant on the basis of the fact that oxidation number in case of oxidant decreases but in case of reductant increases.

Thus, here Cu_2O is oxidant and Cu_2S is reductant.

Quick Trick

- The equivalent weight of oxidising/reducing agent
$$= \frac{\text{molecular weight of the reagent}}{\text{change in oxidation number}}$$
- e.g. In acidic medium, equivalent weight of $KMnO_4$ is calculated as follows

  The reaction in acidic medium is



- Thus, equivalent weight of $KMnO_4$
$$= \frac{\text{molecular weight of } KMnO_4}{\text{change in oxidation number}}$$
$$= \frac{39 + 55 + 4 \times 16}{5} = 31.6$$

Limitations of Concept of Oxidation Number

The limitations of concept of oxidation number are as follows

(i) Many a times the result of concept of oxidation state does not resemble with that obtained from classical approach.
e.g. Consider the following reaction,

$$\overset{+4\ -1}{SiCl_4} + \overset{+1\ +3\ -1}{LiAlH_4} \longrightarrow \overset{+4\ -1}{SiH_4} + \overset{+1\ -1}{LiCl} + \overset{+3\ -1}{AlCl_3}$$

According to classical approach, here silicon is reduced and lithium as well as aluminium are oxidised. But these results are not supported by concept of oxidation number as none of the species is undergoing any change in oxidation number.

(ii) Further it is difficult to classify the following reaction as redox reaction on the basis of concept of oxidation number.

$$CH_3NCO + H_2O \longrightarrow CH_3NH_2 + CO_2$$

In order to overcome these limitations, the concept of redox reactions has been proposed in term of electron density.

A decrease in electron density shows the oxidation process whereas an increase in electron density around the atom(s) involved in the reaction shows a reduction process.

Topic Practice 1

1. Identify the substance that get reduced in the following reaction.

$$Fe_2O_3(s) + 3\,CO\,(g) \longrightarrow 2\,Fe\,(s) + 3\,CO_2(g)$$

2. The compound AgF_2 is unstable compound. However, if formed, the compound acts as a very strong oxidising agent. Why? **NCERT**

Hint:

- Due to the presence of completely filled configuration Ag^+ is more stable than Ag^{2+}.
- An oxidising agent has a great tendency of accepting electron.

3. In the given reaction which substance is getting oxidised and which substance is getting reduced? Name reducing agent and oxidising agent.

$$2\,Cu_2S + 3O_2 \longrightarrow 2\,Cu_2O + 2\,SO_2$$

4. Nitric acid acts only as an oxidising agent while nitrous acid acts both as an oxidising as well as reducing agent. Explain.

5. Why does the following reaction occur?

$$XeO_6^{4-}(aq) + 2F^-(aq) + 6H^+(aq) \longrightarrow$$

$$XeO_3(g) + F_2(g) + 3H_2O\,(l)$$

What conclusion about the compound Na_4XeO_6 (of which XeO_6^{4-} is a part) can be drawn from the reaction? **NCERT**

6. Consider the reactions,

$$2S_2O_3^{2-}(aq) + I_2\,(s) \longrightarrow S_4O_6^{2-}(aq) + 2I^-(aq)$$

$$S_2O_3^{2-}(aq) + 2Br_2(l) + 5H_2O\,(l) \longrightarrow 2SO_4^{2-}(aq) + 4Br^-(aq) + 10H^+(aq)$$

Why does the same reductant, thiosulphate react differently with iodine and bromine? **NCERT**

7. Whenever a reaction between an oxidising agent and a reducing agent is carried out, a compound of lower oxidation state is formed if the reducing agent is in excess and a compound of higher oxidation state is formed if the oxidising agent is in excess. Justify this statement giving three illustrations. **NCERT**

8. Does the oxidation number of an element in any molecule or any polyatomic ion represent the actual charge on it?

9. Assign oxidation number to the underlined elements in each of the following species. **NCERT**

(i) $NaH_2\underline{P}O_4$ (ii) $NaH\underline{S}O_4$ (iii) $H_4\underline{P}_2O_7$

(iv) $K_2\underline{Mn}O_4$ (v) $Ca\underline{O}_2$ (vi) $Na\underline{B}H_4$

(vii) $H_2\underline{S}_2O_7$ (viii) $KAl(\underline{S}O_4)_2 \cdot 12H_2O$

(ix) $\underline{Cr}_2O_7^{2-}$ (x) $\underline{S}_2O_3^{2-}$

10. Consider the elements : Cs, Ne, I and F.

(i) Identify the element that exhibits only negative oxidation state.

(ii) Identify the element that exhibits only positive oxidation state.

(iii) Identify the element that exhibits both positive and negative oxidation states.

(iv) Identify the element which exhibits neither the negative nor does the positive oxidation state. **NCERT**

11. Calculate the oxidation number of each sulphur atom in the following compounds **NCERT Exemplar**

(i) $Na_2S_2O_3$ (ii) $Na_2S_4O_6$

(iii) Na_2SO_3 (iv) Na_2SO_4

12. Zn rod is immersed in $CuSO_4$ solution. What will you observe after an hour? Explain your observation in terms of redox reaction.

13. The compound *Y* $Ba_2Cu_3O_7$, which shows superconductivity, has copper in *x* oxidation state. Assume that the rare earth element yttrium is in its usual +3 oxidation state. Predict the value of *x*.

14. Consider the reactions,

(i) $H_3PO_2(aq) + 4AgNO_3(aq) + 2H_2O\,(l) \longrightarrow H_3PO_4(aq) + 4\,Ag\,(s) + 4HNO_3(aq)$

(ii) $H_3PO_2(aq) + 2\,CuSO_4(aq) + 2H_2O\,(l) \longrightarrow H_3PO_4(aq) + 2Cu\,(s) + H_2SO_4(aq)$

(iii) $C_6H_5CHO(l) + 2\,[Ag(NH_3)_2]^+(aq) + 3OH^-(aq) \longrightarrow C_6H_5COO^-(aq) + 2Ag\,(s) + 4\,NH_3(aq) + 2H_2O\,(l)$

(iv) $C_6H_5CHO\,(l) + 2Cu^{2+}(aq) + 5OH^-(aq) \longrightarrow$ No change observed.

What inference do you draw about the behaviour of Ag^+ and Cu^{2+} from these reactions? **NCERT**

Hint: To find the behaviour of Ag^+ and Cu^{2+} in the given reactions, find whether there is addition of oxygen or any other electronegative element and addition of hydrogen or any other electropositive element. If there is addition of O to a reactant then other reactant acts as reducing agent (or reductant) and if there is addition of hydrogen, they act as oxidant.

15. **Why do the following reactions proceed differently?**

$Pb_3O_4 + 8\,HCl \longrightarrow 3\,PbCl_2 + Cl_2 + 4H_2O$ and

$Pb_3O_4 + 4\,HNO_3 \longrightarrow 2Pb(NO_3)_2 + PbO_2 + 2H_2O$

NCERT

16. **Decide whether each of the following reaction involves oxidation-reduction reaction or not. If it does, identify which species is oxidised and which gets reduced?**

(i) $4\,CH_3\overset{\overset{O}{\|}}{C}CH_3 + LiAlH_4 + 4H_2O \rightleftharpoons 4\,CH_3\overset{\overset{OH}{|}}{C}HCH_3 + LiOH + Al(OH)_3$

(ii) $CH_3CH_2OH \xrightarrow{H_2SO_4} CH_2{=}CH_2 + H_2O$

(iii) $CH_3\overset{\overset{O}{\|}}{C}OH + CH_3NH_2 \rightleftharpoons CH_3\overset{\overset{O}{\|}}{C}O^- + CH_3NH_3^+$

Redox Reactions : Types, Balancing and Applications

Types of Redox Reactions

The redox reactions can be classified into following types

Combination Reactions

The reactions in which two or more atoms or molecules combine together to give only a single compound are called combination reactions. All combustion reactions are included in this category.

e.g. $\overset{0}{2Mg}\,(s) + \overset{0}{O_2}\,(g) \longrightarrow \overset{+2\ -2}{2MgO}(s)$

For a combination reaction to be a redox reaction atleast one of its reactant must be in its elemental form. All combustion reactions, which make use of elemental dioxygen, as well as other reactions involving elements other than dioxygen are called redox reactions.

$$\overset{0}{C}(s) + \overset{0}{O_2}\,(g) \xrightarrow{\Delta} \overset{+4\ -2}{CO_2}\,(g)$$

$$\overset{0}{3Mg}\,(s) + N_2\,(g) \xrightarrow{\Delta} \overset{+2\ \ -3}{Mg_3N_2}\,(g)$$

Decomposition Reactions

These are just opposite to the combination reaction, i.e. in these reactions, one compound split into two or more simpler compounds, atleast one of which must be in elemental form.

e.g. $\overset{+\,1\,-\,1}{2NaH}\,(s) \xrightarrow{\Delta} \overset{0}{2Na}\,(s) + \overset{0}{H_2}\,(g)$

$$\overset{+2\,+4\,-2}{CaCO_3}\,(s) \xrightarrow{\Delta} \overset{+2\ -2}{CaO}(s) + \overset{+4\ -2}{CO_2}\,(g)$$

The former reaction involves change in oxidation states. So, it is considered as redox reaction.

The latter reaction, i.e. the decomposition of calcium carbonate is not a redox reaction as there is no change in the oxidation states of the elements involving in the reaction. It suggests that **all the decomposition reactions are not the redox reactions.**

Displacement Reactions

The reactions in which an atom (or ion) replaces the ion (or atom) of another element from a compound, are known as displacement reactions. These reactions can be represented as

$$X + YZ \longrightarrow XZ + Y$$

Depending upon the nature of element displaced, these reactions can be classified into two categories.

1. Metal Displacement Reactions

When a free metal displaces another metal from its compounds, the reaction is known as metal displacement reactions. In general, a more reactive metal displaces the less reactive one from its compounds.

e.g. $\overset{+2+6\ -2}{CuSO_4}\,(aq) + \overset{0}{Zn}(s) \longrightarrow \overset{0}{Cu}(s) + \overset{+2\ +6\ -2}{ZnSO_4}\,(aq)$

$$\overset{+5\ -2}{V_2O_5}\,(s) + \overset{0}{5Ca}\,(s) \xrightarrow{\Delta} \overset{0}{2V}(s) + 5\,\overset{+2\ -2}{CaO}(s)$$

In the above reactions, the metal which displaces the other from their salt is a good reducing agent, i.e. have more tendency to lose electrons as compared to the one that is displaced or reduced. Above reactions also reveals that the metal displacement reactions are generally redox reactions. An important application of these reactions is in metallurgical processes where pure metals are obtained from their ores.

2. Non-metal Displacement Reactions

When some of the non-metal is displaced by a metallic or a non-metallic element, the reaction is said to be non-metal displacement reaction.

In these reactions, generally the hydrogen is displaced but in some reactions, displacement of oxygen or halogen has also been observed.

(i) **Reactions in which hydrogen is displaced** Examples of these reactions are as follows

(a) Highly reactive metals like alkali metals and some alkaline earth metals (such as Ca, Sr and Ba) being good reductants or reducing agents, displace hydrogen from cold water.

$$\text{e.g. } \overset{0}{2Na}(s) + \underset{\text{(Cold)}}{\overset{+1-2}{2H_2O}}(l) \longrightarrow \overset{+1-2+1}{2NaOH}(aq) + \overset{0}{H_2}(g)$$

$$\overset{0}{Ca}(s) + \underset{\text{(Cold)}}{\overset{+1-2}{2H_2O}}(l) \longrightarrow \overset{+2-2+1}{Ca(OH)_2}(aq) + \overset{0}{H_2}(g)$$

(b) Less reactive metals like magnesium and iron produce hydrogen only when they are treated with steam.

$$\overset{0}{Mg}(s) + \underset{\text{(Steam)}}{\overset{+1-2}{2H_2O}}(l) \xrightarrow{\Delta} \overset{+2-2+1}{Mg(OH)_2}(s) + \overset{0}{H_2}(g)$$

$$\overset{0}{2Fe}(s) + \underset{\text{(Steam)}}{\overset{+1-2}{3H_2O}}(l) \xrightarrow{\Delta} \overset{+3-2}{Fe_2O_3}(s) + \overset{0}{3H_2}(g)$$

(c) Several metals including those which do not displace hydrogen from cold water or even with steam can displace hydrogen from mineral acids (i.e. HCl, H_2SO_4, etc.).

$$\text{e.g. } \overset{0}{Zn}(s) + \overset{+1-1}{2HCl}(aq) \longrightarrow \overset{+2-1}{ZnCl_2}(aq) + \overset{0}{H_2}(g)$$

Note Highly reactive metals (e.g. Mg) readily lose electrons and get oxidised. These electrons are accepted by the H^+ ions provided by mineral acid. After accepting electrons H^+ ions get converted into hydrogen gas. Thus, hydrogen gas is evolved. However, least reactive metals because of their less tendency to lose electron, are generally unable to convert H^+ into H_2. Thus, there is no evolution of hydrogen with less reactive metals.

(ii) **Reaction in which oxygen is displaced** Fluorine being the strongest oxidising agent displaces oxygen when treated with water.

$$\text{e.g. } \overset{+1-2}{2H_2O}(l) + \overset{0}{2F_2}(g) \longrightarrow \overset{+1-1}{4HF}(aq) + \overset{0}{O_2}(g)$$

(iii) **Reactions in which halogens are displaced** Fluorine has maximum oxidising power and can oxidise chlorine, bromine and iodine. However, because of its high reactivity with water, its reaction with chloride, bromide and iodide are not generally carried out in aqueous solution.

Chlorine, however, is capable of displacing bromide and iodide ions from their aqueous solution.

$$\overset{0}{Cl_2}(g) + \overset{+1-1}{2KBr}(aq) \longrightarrow \overset{+1-1}{2KCl}(aq) + \overset{0}{Br_2}(l)$$

(Oxidation)

$$\overset{0}{Cl_2}(g) + \overset{+1-1}{2KI}(aq) \longrightarrow \overset{+1-1}{2KCl}(aq) + \overset{0}{I_2}(s)$$

Bromine can displace only iodide ion from its solution.

$$\overset{0}{Br_2}(l) + \overset{-1}{2I^-}(aq) \longrightarrow \overset{-1}{2Br^-}(aq) + \overset{0}{I_2}(s)$$

The halogen displacement reactions have a direct industrial application.

Disproportionation Reactions

These are a special kind of redox reactions in which the same element undergoes oxidation as well as reduction.

In these reaction, one of the reacting substances always contains an element that can exist in at least three oxidation states. The element present in the reacting substance, is in the intermediate oxidation state and both higher and lower oxidation states of that element are formed in the product. e.g. In hydrogen peroxides, disproportionation reaction occurs as

$$\overset{+1\ -1}{2H_2O_2}(aq) \longrightarrow \overset{+1\ -2}{2H_2O}(l) + \overset{0}{O_2}(g) \quad \ldots(i)$$

(Reduction; Oxidation)

Similarly, chlorine, sulphur and phosphorus undergo disproportionation in the alkaline medium as shown below.

$$\overset{0}{Cl_2}(g) + 2OH^-(aq) \longrightarrow \overset{+1}{ClO^-}(aq) + \overset{-1}{Cl^-}(aq) + H_2O(l) \quad \ldots(ii)$$

$$\overset{0}{S_8}(s) + 12OH^-(aq) \longrightarrow 4S^{2-}(aq) + 2\overset{+2}{S_2}O_3^{2-}(aq) + 6H_2O(l) \quad \ldots(iii)$$

$$P_4(s) + 3OH^-(aq) + 3H_2O(l) \longrightarrow \overset{-3}{PH_3}(g) + 3H_2\overset{+1}{PO_2^-}(aq) \quad \ldots(iv)$$

The (ClO^-) formed in the reaction (ii) oxidises the colour-bearing stains of the substances to coloureless substances.

Also, bromine and iodine show same trend as shown by chlorine in reaction (ii).

Whereas fluorine shows deviation from this behaviour when reacts with alkali. The reaction involve in case of fluorine is shown below.

$$2F_2(g) + 2OH^-(aq) \longrightarrow 2F^-(aq) + OF_2(g) + 2H_2O(l) \quad \ldots(v)$$

No positive oxidation states of fluorine.

fluorine also react with water to produce some oxygen. This behaviour shown by fluorine is not surprising as we known that fluorine do not exhibit any positive oxidation state. Thus, fluorine does not show a disproportionation tendency.

EXAMPLE 8. Suggest a scheme of classification of the following redox reactions.

(i) $N_2(g) + O_2(g) \longrightarrow 2NO(g)$

(ii) $2Pb(NO_3)_2(s) \longrightarrow 2PbO(s) + 2NO_2(g) + \frac{1}{2}O_2(g)$

(iii) $NaH(s) + H_2O(l) \longrightarrow NaOH(aq) + H_2(g)$

(iv) $2NO_2(g) + 2OH^-(aq) \longrightarrow NO_2^-(aq) + NO_3^-(aq) + H_2O(l)$

NCERT

Sol. (i) Since, the two molecules are combining to give a single product, so it is an example of **combination reaction.**

(ii) Since, the compound splits up into several simpler substances, out of which O_2 is in the elemental form, thus, it is an example of **decomposition redox reaction.**

(iii) $NaH(s) + H_2O(l) \longrightarrow NaOH(aq) + H_2(g)$

The ionic form of the reaction is

$$H^-(s) + H_2O(l) \longrightarrow OH^-(aq) + H_2(g)$$

Here, H^- (hydride ion) displaces hydrogen from water molecule, so it is an example of **displacement redox reaction.**

(iv) $\overset{-4}{2NO_2}(g) + 2OH^-(aq) \longrightarrow \overset{+3}{NO_2^-}(aq) + \overset{+5}{NO_3^-}(aq) + H_2O(l)$

Oxidation

Reduction

Since, in the above reaction, the same element undergoes oxidation as well as reduction, so it is an example of **disproportionation reaction.**

EXAMPLE 9. Which of the following species, do not show disproportionation reaction and why?

ClO^-, ClO_2^-, ClO_3^- and ClO_4^-

Also write reaction for each of the species that disproportionates.

NCERT

Sol. Disproportionation reaction is shown by the species in which the element exhibit intermediate oxidation state. Among the oxo anions of chlorine listed above, ClO_4^- does not disproportionate because in the oxo anion chlorine is present in its highest oxidation state that is +7. The disproportionation reactions for the other three oxoanions of chlorine are as follows

$$\overset{+1}{3ClO^-} \longrightarrow \overset{-1}{2Cl^-} + \overset{+5}{ClO_3^-}$$

$$\overset{+3}{6ClO_2^-} \xrightarrow{h\nu} \overset{+5}{4ClO_3^-} + \overset{-1}{2Cl^-}$$

$$\overset{+5}{4ClO_3^-} \longrightarrow \overset{-1}{Cl^-} + \overset{+7}{3ClO_4^-}$$

Balancing of Redox Reactions

In order to satisfy the law of conservation of mass (which states that matter or mass can neither be created nor destroyed), the chemical reactions are balanced, i.e. the number of atoms of each element equalise on both sides of the chemical equation.

Several methods can be used to balance a redox reaction. These methods are as follows

(i) Hit and Trial method
(ii) Partial equation method
(iii) Oxidation number method
(iv) Half equation method or Ion-electron method.

[All these methods are in use and the choice of their use depends upon the individual using them.]

The former two methods have been described in chapter 1. The discussion of the later two methods is given below.

Oxidation Number Method

This method is based on the change in the oxidation number of oxidant and reductant. The common steps involves in this method is as follows

Step I **Writing skeleton ionic equation** Write the correct formula for each reactant and product and draw an arrow in between them.

Convert it into skeleton ionic equation by removing the spectator ion, the ion which is not involve in the reaction.

Step II **Assigning oxidation state to each element and balancing atoms other than H and O** Write the oxidation number of each element above its symbol and balance all the atoms other than H and O.

Step III **Finding change in oxidation number and equalise it** Calculate the decrease or increase in the oxidation number per atom and for the entire molecule/ion in which it is present. If the decrease and increase in oxidation number is not equal, then equalise them by multiplying with a suitable number.

Note If in any equation, you find that all the reactants are only oxidised or reduced then there is something wrong. It may be the formula of reactants or products or the oxidation numbers of the element have not been assigned properly.

Step IV **Balancing ionic charge by adding H^+ or OH^-** In case of reactions occurring in water, add H^+ or OH^- ions on the appropriate side to balance the total ionic charges of reactants and products.

If medium is acidic, use H^+ ions and if medium is basic, use OH^- ions on the appropriate side.

Step V **Balancing H and O-atoms** In order to equalise the number of hydrogen atoms, add water (H_2O) molecules on appropriate side.

Now, check the number of oxygen atoms on both the sides. If these are found equal, then the equation is a balanced chemical equation.

EXAMPLE 10. Write the net ionic equation for the reaction of potassium dichromate (VI), $K_2Cr_2O_7$ with sodium sulphite, Na_2SO_3, in an acid solution to give chromium (III) ion and the sulphate ion **NCERT**

Sol. Following steps are involved in order to solve out such problems.

Step I Writing skeletal ionic equation.

e.g.

$$Cr_2O_7^{2-}(aq) + SO_3^{2-}(aq) \longrightarrow Cr^{3+}(aq) + SO_4^{2-}(aq)$$

Step II Assigning oxidation number to each element and balancing atoms other than H and O.

$$\overset{+6}{Cr_2}\overset{-2}{O_7^{2-}}(aq) + \overset{+4}{S}\overset{-2}{O_3^{2-}}(aq) \longrightarrow \overset{+3}{Cr^{3+}}(aq) + \overset{+6}{S}\overset{-2}{O_4^{2-}}(aq)$$

Step III Finding change in oxidation number and equalise it.

Change in O.N. = 6 − 4 = 2] × 3

$$\overset{+6}{Cr_2}\overset{-2}{O_7^{2-}}(aq) + \underline{3}\overset{+4}{S}\overset{-2}{O_3^{2-}}(aq) \longrightarrow \underline{2}\overset{+3}{Cr^{3+}}(aq) + \underline{3}\overset{+6}{S}\overset{-2}{O_4^{2-}}(aq)$$

Change in O.N. = 6 − 3 = 3 per atom

or 3 × 2 = 6 net change (as two Cr atoms are involved)

To equalise the increase in oxidation state of S with decrease in oxidation state of Cr, we have to multiply it with 3, thus we have to multiply SO_3^{2-} and SO_4^{2-} with 3.

Step IV **Balancing ionic charge**

Since, the medium is acidic, so we can use only H^+ ions to balance the charge.

$$Cr_2O_7^{2-}(aq) + 3SO_3^{2-}(aq) \longrightarrow 2Cr^{3+}(aq) + 3\,SO_4^{2-}(aq)$$

$= (-2) + (-2 \times 3) = -8$ $\qquad$ $= (2 \times +3) + (-2 \times 3) = 0$

So, to balance −8 charge, we have to add $8H^+$ ions at the reactant side.

Thus, the reaction becomes

$$Cr_2O_7^{2-}(aq) + 3SO_3^{2-}(aq) + 8H^+(aq) \longrightarrow 2Cr^{3+}(aq) + 3\,SO_4^{2-}(aq)$$

Step V **Balancing of H and O-atoms**

Now count the total number of H-atoms on both the sides and add H_2O molecules to equalise them.

In the above reaction, there are 8H-atoms towards the reactant side, so in order to balance them, we add $4H_2O$ towards the product side.

$$Cr_2O_7^{2-}(aq) + 3SO_3^{2-}(aq) + 8H^+(aq) \longrightarrow 2Cr^{3+}(aq) + 3\,SO_4^{2-}(aq) + 4\,H_2O$$

Now, we check the number of O-atoms. Since, the number of O-atoms are already balanced, so the equation is a balanced chemical equation.

EXAMPLE 11. Permanganate ion reacts with bromide ion in basic medium to give manganese dioxide and bromate ion. Write the balanced ionic equation for the reaction. **NCERT**

Sol. Step I The skeletal ionic equation is

$$MnO_4^-(aq) + Br^-(aq) \longrightarrow MnO_2(s) + BrO_3^-(aq)$$

Step II Assign oxidation numbers for Mn and Br

$$\overset{+7}{Mn}\,O_4^-(aq) + \overset{-1}{Br^-}(aq) \longrightarrow \overset{+4}{Mn}\,O_2(s) + \overset{+5}{Br}O_3^-(aq)$$

This indicates that permanganate ion is the oxidant and bromide ion is the reductant.

Step III Calculate the increase and decrease of oxidation number, and make the increase equal to the decrease.

$$2\overset{+7}{Mn}\,O_4^-(aq) + \overset{-1}{Br^-}(aq) \longrightarrow 2\,\overset{+4}{M}\,nO_2(s) + \overset{+5}{Br}O_3^-(aq)$$

Step IV As the reaction occurs in the basic medium, and the ionic charges are not equal on both sides, add $2OH^-$ ions on the right to make ionic charges equal.

$$2MnO_4^-(aq) + Br^-(aq) \longrightarrow 2MnO_2(s) + BrO_3^-(aq) + 2OH^-(aq)$$

Step V Finally, count the hydrogen atoms and add appropriate number of water molecules (i.e. one H_2O molecule) on the left side to achieve balanced redox change.

$$2MnO_4^-(aq) + Br^-(aq) + H_2O(l) \longrightarrow 2MnO_2(s) + BrO_3^-(aq) + 2OH^-(aq)$$

Half Reaction Method or Ion Electron Method

This method is based upon the splitting of a redox reaction into two half reactions (one is oxidation half reaction and the other is reduction half reaction) and includes the following steps.

Step I **Converting unbalanced equation into ionic form** Break all the ionic compounds involved in the reaction into their constituent ions and cancel out the spectator ion to convert a molecular reaction into ionic form.

Step II **Splitting the equation into two halves** Separate the equation into oxidation half and reduction half reactions.

Step III **Balancing all elements except H and O** Balance all other elements except H and O in each half reaction individually.

Step IV **Balancing H and O-atoms** In acidic medium, add H_2O to balance O-atoms and H^+ to balance hydrogen atoms. However, if the medium is basic, add OH^- equal to number of H^+ ions on both the sides of the equation.

Step V **Balancing the charge**

(i) Balance the charge in each half equation by adding electrons toward the side having more positive charge. (The number of electrons added should be equal to the excess in positive charge).

(ii) Now, multiply the half equations by suitable integer so that the number of electrons (charge) become equal in both the half equations.

Step VI Adding the half equations Add the two half equations and cancel the electrons on each side to get the overall ionic equation.

Step VII Verification of balanced equation Check the correctness of equation by comparing the number of atoms of each element on both the sides and also by comparing the total charges of both the sides.

Step VIII Conversion of ionic equation into molecular form Convert the obtained equation into molecular form by adding spectator ions. (If mentioned in the question).

EXAMPLE 12. In acidic medium, dichromate ions oxidise ferrous sulphate to ferric sulphate and itself gets reduced to Cr^{3+} ions. Write a balanced equation to represent this redox reaction. (Use ion electron method). **NCERT**

Sol. The equation for the given reaction is

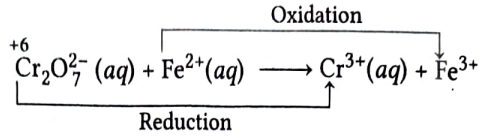
$$Cr_2O_7^{2-}(aq) + FeSO_4 \longrightarrow Cr^{3+} + Fe_2(SO_4)_3$$

Balancing of this equation by ion electron method involves the following steps.

Step I **Converting unbalanced equation into ionic form** It can be achieved by splitting the ionic compounds into constituent ions and by cancelling spectator ions.

$$Cr_2O_7^{2-}(aq) + Fe^{2+}(aq) + SO_4^{2-}(aq) \longrightarrow Cr^{3+}(aq) + Fe^{3+}(aq) + SO_4^{2-}(aq)$$

Here, SO_4^{2-} is the spectator ion so after leaving it, the reaction becomes



Step II **Splitting the equation into two halves**

Oxidation half equation, $\overset{+2}{Fe^{2+}}(aq) \longrightarrow \overset{+3}{Fe^{3+}}(aq)$

Reduction half equation, $\overset{+6}{Cr_2}O_7^{2-}(aq) \longrightarrow \overset{+3}{Cr^{3+}}(aq)$

Step III **Balancing all elements other than H and O**

Here, oxidation half reaction is already balanced with respect to Fe atoms but to balance reduction half reaction, we have to multiply the Cr^{3+} by 2.

$$Cr_2O_7^{2-}(aq) \longrightarrow 2Cr^{3+}(aq)$$

Step IV **Balancing H and O-atoms** In oxidation half reaction, there is no need to balance H and O atoms, as in it these are not present. In reduction half reaction, we have to add $7H_2O$ towards product side to balance O-atoms and then add $14H^+$ towards reactant side to balance H-atoms.

$$Cr_2O_7^{2-} + \underset{\text{To balance H-atoms}}{14H^+} \longrightarrow 2Cr^{3+} + \underset{\text{To balance O-atoms}}{7H_2O}$$

Step V **Balancing of charge**

(i) $\overset{+2}{Fe^{2+}}(aq) \longrightarrow \overset{+3}{Fe^{3+}}(aq)$

So, we have to add one electron towards Fe^{3+} side.

Thus, $Fe^{2+}(aq) \longrightarrow Fe^{3+}(aq) + e^-$

Similarly, $Cr_2O_7^{2-}(aq) + 14H^+(aq) \longrightarrow 2Cr^{3+}(aq) + 7H_2O(l)$

$-2 + 14 \qquad 2 \times (+3)$

$+12 \qquad +6$

$+6 \qquad 0$

So, we have to add $6e^-$ towards the reactant side.

$$Cr_2O_7^{2-}(aq) + 14H^+(aq) + 6e^- \longrightarrow 2Cr^{3+}(aq) + 7H_2O(l)$$

(ii) $[Fe^{2+}(aq) \longrightarrow Fe^{3+}(aq) + e^-] \times 6$ to equalise number of electrons.

Step VI **Adding the half reactions** On adding the half reaction, we get the overall balanced equation.

$$6Fe^{2+}(aq) + Cr_2O_7^{2-}(aq) + 14H^+(aq) \longrightarrow 6Fe^{3+}(aq) + 2Cr^{3+}(aq) + 7H_2O(l)$$

Step VII **Verification of balanced equation** Since, the number of atoms of each element are equal on both the sides, so it is a balanced chemical equation.

Further:

$$6\,Fe^{2+}(aq) + Cr_2O_7^{2-}(aq) + 14\,H^+(aq) \longrightarrow 6\,Fe^{3+}(aq) + 2\,Cr^{3+}(aq) + 7\,H_2O\,(l)$$

$$6\times(+2) + (-2) + 14\,(+1) \equiv 6\times(+3) + 2\times(+3)$$

$$12 - 2 + 14 \equiv 18 + 6 \Rightarrow 24 = 24$$

Thus, the equation is also balanced with respect to charge.

Step VIII Conversion of ionic equation into molecular form

As we know SO_4^{2-} is the spectator ion, and there are 6 Fe^{2+}(which require 6 SO_4^{2-} ions) and 14H^+ ions (that require 7 SO_4^{2-} ions), so, we have to add 13 SO_4^{2-} ions on both the sides.

$$6\,Fe^{2+}(aq) + 6\,SO_4^{2-}(aq) + Cr_2O_7^{2-}(aq) + 14H^+(aq) + 7SO_4^{2-}(aq) \longrightarrow 6\,Fe^{3+}(aq) + 9\,SO_4^{2-}(aq) + 2Cr^{3+} + 3\,SO_4^{2-}(aq) + 7\,H_2O + SO_4^{2-}$$

or $$6\,FeSO_4 + Cr_2O_7^{2-}(aq) + 7H_2SO_4 \longrightarrow 3\,Fe_2(SO_4)_3 + Cr_2(SO_4)_3 + 7H_2O + SO_4^{2-}$$

Applications of Redox Reactions

Redox reactions are widely used in extraction of metals, electrochemical cells, titrations, quantitative analysis. In this section we will study titrations and electrode processes.

Redox Reactions as the Basis for Titration

Before discussing the role of redox reaction in titrations, it is necessary to became aware from the basic terminology used in titrimetric exercises.

Terminology Used in Titrimetric Exercises

The main terms used in titrimetric exercises are discussed below.

(i) **Titration** It is a common laboratory technique used to calculate the strength of unknown solution quantitatively against a standard solution, using a colour/pH sensitive substance.

(ii) **Standard solution or titrant** It is the solution of known concentration and is taken in the burette during titrimetric exercises.

(iii) **Analyte** This term is used for solution of unknown concentration taken in titration flask.

(iv) **End point or equivalence point** The stage at which the two substance (standard solution and solution of unknown concentration) react completely is called the equivalence point or end point. It is determined with the help of indicator.

(v) **Indicator** It is a colour/pH sensitive substance, used to find the end point of a titration.

Redox Titrations

In redox titrations, the strength of a reductant/oxidant is determined by titrating it with standard solution of oxidant/reductant using a redox sensitive indicator. The indicators used for redox titrations depend upon the nature of redox reaction or reagent and product involved. Some examples of indicators used in different conditions of redox titrations are given below.

(i) When One of the Reagents is Intensely Coloured

In redox titrations, if one of the reagents is intensely coloured, then it itself behaves as an indicator. Such an indicator is called self or auto-indicator, e.g. permanganate ion (MnO_4^-). Reactions involved in these titrations are

$$MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O$$

$$[Fe^{2+} \longrightarrow Fe^{3+} + e^-]\times 5$$

$$MnO_4^- + 8H^+ + 5Fe^{2+} \longrightarrow 5Fe^{3+} + Mn^{2+} + 4H_2O$$

(ii) When One of the Reagents Oxidises the Indicator

These are the substances that get oxidised immediately after the consumption of last drop of reductant and produce a drastic colour change. It is generally the case of potassium dichromate ($K_2Cr_2O_7$) titrations.

The indicators used in its titrations are of two types

(a) **External indicators** These indicators are not added inside the reaction mixture but after the equivalence point, drops of indicator is placed over few drops of reaction mixture taken in watch glass with the help of glass rod to produce an intense blue colour, thus indicating the end point. Potassium ferricyanide is an example of such indicator.

(b) **Internal indicators** These are added directly into the solution or titration flask before the process of titration. Diphenylamine (which produces intense blue colour at the end point) and N-phenylanthranilic acid (which produces violet-red colour at the end point) are the examples of such indicators.

(iii) When the Reagent can Oxidise Iodide Ion into Iodine or Iodine to Iodide Ion

(a) Titration in which the liberated iodine (from an oxidising agent) is titrated with standard solution of sodium thiosulphate is called iodometric titration.

e.g. $2Cu^{2+}(aq) + 4I^-(aq) \longrightarrow Cu_2I_2(s) + I_2(aq)$

(in presence of starch forms blue)

$$2Na_2S_2O_3(aq) + I_2(aq) \longrightarrow 2NaI + Na_2S_4O_6(aq)$$

Blue colour — Blue colour disappear

(b) When iodine is titrated directly against a reducing agent such as sodium thiosulphate, the titration is known as **iodimetric titration.**

e.g. $I_2(aq) + 2S_2O_3^{2-}(aq) \longrightarrow 2I^-(aq) + S_4O_6^{2-}(aq)$

Blue with starch; Blue colour disappear; Tetrathionate ion

n such reactions **starch** acts as the indicator as it roduces violet or intense blue colour with the molecular odine, so helps in detection of end point.

Stoichiometry of Redox Reactions in Solutions

Stoichiometry of redox reactions means the amounts of reductants and oxidants in redox reactions. These calculations are based on mole concept or molarity equation. (Both of which have been discussed in unit I). For both the methods a balanced redox equation is required.

Recall the molarity equation $\frac{M_1V_1}{n_1} = \frac{M_2V_2}{n_2}$ (Oxidant) (Reductant)

where, M_1 and V_1 are the molarity and volume of oxidant and M_2 and V_2 are the molarity and volume of reductant. n_1 and n_2 are their stoichiometric coefficients.

Redox Reactions and Electrode Processes

The redox reactions in which the oxidation and reduction half reactions occur in the same vessel are called the **direct redox reactions.**

e.g. Placing a zinc strip inside the copper sulphate solution, where Zn is oxidised to Zn^{2+} and copper sulphate is reduced to metallic copper. During this reaction heat is evolved.

The equation of reaction can be represented as shown below

$$Zn(s) + Cu^{2+}(aq) \longrightarrow Zn^{2+}(aq) + Cu(s)$$

The redox reactions in which the oxidation and reduction half reactions are carried out into separate vessels, are called **indirect redox reactions.** In these reactions, the reaction occurs in both the vessels as well as at the interface of the metal. These reactions form the basis of **electrochemical cell** where indirect redox reaction is used as a source of energy and **electrolytic cell,** where a non-spontaneous reaction is carried out with the help of energy.

Terminology Used in Electrode Processes

Before discussing the role of redox reactions in the electrode processes, it is essential to become aware from the following terms.

(i) **Electrodes** A metal rod or strip dipped in its own salt solution is called an **electrode**. e.g. When copper rod is dipped in the copper sulphate solution, then term electrode is used for this rod.

(ii) **Anode** The electrode at which oxidation, i.e. loss of electrons from the metal takes place is called the **anode.** It is negative pole in case of an electrochemical cell but a positive pole in case of an electrolytic cell.

(iii) **Cathode** It is the electrode at which reduction, i.e. gain of electron by the metal ion occurs. It is the positive pole of an electrochemical cell whereas negative pole of an electrolytic cell.

(iv) **Redox couple** The combination of the oxidised and reduced forms of a substance taking part in an oxidation or reduction half reaction is called the **redox couple.** e.g. Zn is the reduced form of Zn^{2+} and Zn^{2+} is the oxidised form of Zn. When these two are present together, they show a redox couple.

A redox couple is represented by separating the oxidised and reduced forms with the help of a vertical line or a slash which shows the interface. e.g. The redox couple of Zn and Zn^{2+} is represented as Zn^{2+}/Zn.

Always remember that in a redox couple the oxidised form is placed before the reduced form. Just like zinc couple, the redox couple for copper is represented as Cu^{2+}/Cu.

(v) **Electrode potential** The potential associated with an electrode is known as the **electrode potential.** It is represented by the symbol E.

(vi) **Standard electrode potential** The potential of an electrode in which the concentration of each species taking part in the electrode reaction is unity (if any gas is involved in the electrode reaction, its pressure is taken as 1 atmosphere) and temperature is maintained at 298 K is called the standard electrode potential. It is represented by $E°$. By convention, the standard electrode potential ($E°$) of hydrogen electrode is 0.00 volts.

Standard electrode potential value of an electrode is indeed the direct measure of relative tendency of the active species in the process to remain in the oxidised/reduced form.

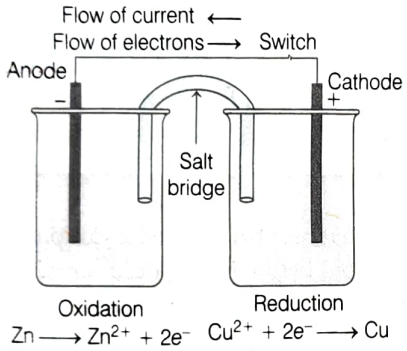
The negative value of $E°$ shows that the electrode has a higher tendency to lose electrons, i.e. acts as a **stronger reducing agent** as compared to a H^+/H_2 couple. Similarly, a positive value of $E°$ shows its higher tendency to gain electrons, i.e. to act as a **stronger oxidising agent** or a weaker reducing agent as compared to H^+/H_2 couple.

Electrochemical Cell

It is a device which is used to convert chemical energy into electrical energy. Daniell or voltaic cell is an important example of this cell.

In order to construct a Daniell cell, zinc rod is placed in 1M solution of zinc sulphate and copper rod is placed in 1M solution of copper sulphate. The beakers containing these two solutions are put side by side and connected through a **salt-bridge.** (A U-tube containing a solution of potassium chloride or ammonium nitrate usually solidified by boiling with *agar-agar* and later cooling to a jelly like substance).

The metallic rods are connected through a wire having provision for a switch. The set up is shown in the following figure.



The set up for Daniell cell

When the switch is in off position, there is no reaction in either of the beakers and no current flows through the metallic wire. When switch is turned on,

- the electrons generated at the anode and reach at the cathode.
- deflection in the galvanometer, indicates the flow of electrons through the connecting wires. The direction of deflection indicates the flow of convention current from copper to zinc.
- the electricity flows from the solution of one beaker to the solution of other beaker through the salt bridge.
- the current flows because the potentials of two redox couples are different.
- the difference between the potentials of two redox couple is called the **cell potential**. It is calculated by using the expression,

$$E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode} = E^\circ_{reduction} + E^\circ_{oxidation}$$

For a cell reaction to be feasible, the E°_{cell} must be positive.

- The copper rod gains mass whereas the zinc rod loses mass.
- The concentration of copper sulphate solution decreases.
- The solutions in the two beakers remain electrically neutral throughout.
- An electrochemical cell stops working after some time. This is because, as cell works, the concentration of metal ion towards the anode increases which results in increased electrode potential.
- At the same time, the concentration of metal ions towards the cathode decreases which results in decrease of electrode potential. Ultimately, a stage is reached at which the electrode potentials of the two half couples become equal, i.e. the cell potential becomes zero, and the cell stops working.

Functions of Salt-Bridge

The main functions of the salt bridge are as follows

- It completes the cell circuit by connecting the solutions of two half cells, without allowing them to mix with each other.
- It maintains the electrical neutrality of the solutions in the two half cells because of the flow of ions.
- The salt-bridge contains only those electrolytes for which the ionic mobilities of cations and anions are nearly the same.
- If salt-bridge is removed, the cell potential drops to zero, i.e. there is no current flow. This is because of the accumulation of positive ions around the anode and negative ions (that are obtained after the reduction of metal ion) around the cathode, which results in the development of charges in the solution and hence, the current stops flowing.

Representation of an Electrochemical Cell

To represent an electrochemical cell, the anode half reaction (or electrode) is written at the left and cathode half reaction (or electrode) is written at the right. These two are separated by two vertical lines representing the salt bridge.

Thus, a Daniell cell can be represented as

Anode half reaction (oxidation)

$$Zn \longrightarrow Zn^{2+} + 2e^-$$

Cathode half reaction (reduction)

$$Cu^{2+} + 2e^- \longrightarrow Cu$$

Thus, the Daniell cell is $\underset{\text{Anode}}{Zn|Zn^{2+}} \quad \underset{\text{Salt-bridge}}{||} \quad \underset{\text{Cathode}}{Cu^{2+}|Cu}$

[More details about the representation of cell will be discussed in electrochemistry chapter of Class XII].

Electrochemical Series

When different electrodes or redox couples are arranged in decreasing order of their standard reduction potentials, E° (or standard electrode potentials), the obtained series is called **electrochemical series** or **electromotive series**. The series is given below.

The Electrochemical Series (E° values at 298 K)

Reaction (Oxidised form + ne^- ⟶ Reduced form)		E°/V
$F_2(g) + 2e^-$	$\rightarrow 2F^-$	2.87
$Co^{3+} + e^-$	$\rightarrow Co^{2+}$	1.81
$H_2O_2 + 2H^+ + 2e^-$	$\rightarrow 2H_2O$	1.78
$MnO_4^- + 8H^+ + 5e^-$	$\rightarrow Mn^{2+} + 4H_2O$	1.51
$Au^{3+} + 3e^-$	$\rightarrow Au(s)$	1.40
$Cl_2(g) + 2e^-$	$\rightarrow 2Cl^-$	1.36
$Cr_2O_7^{2-} + 14H^+ + 6e^-$	$\rightarrow 2Cr^{3+} + 7H_2O$	1.33
$O_2(g) + 4H^+ + 4e^-$	$\rightarrow 2H_2O$	1.23
$MnO_2(s) + 4H^+ + 2e^-$	$\rightarrow Mn^{2+} + 2H_2O$	1.23
$Br_2 + 2e^-$	$\rightarrow 2Br^-$	1.09
$NO_3^- + 4H^+ + 3e^-$	$\rightarrow NO(g) + 2H_2O$	0.97
$2Hg^{2+} + 2e^-$	$\rightarrow Hg_2^{2+}$	0.92
$Ag^+ + e^-$	$\rightarrow Ag(s)$	0.80
$Fe^{3+} + e^-$	$\rightarrow Fe^{2+}$	0.77
$O_2(g) + 2H^+ + 2e^-$	$\rightarrow H_2O_2$	0.68
$I_2(s) + 2e^-$	$\rightarrow 2I^-$	0.54
$Cu^+ + e^-$	$\rightarrow Cu(s)$	0.52
$Cu^{2+} + 2e^-$	$\rightarrow Cu(s)$	0.34
$AgCl(s) + e^-$	$\rightarrow Ag(s) + Cl^-$	0.22
$AgBr(s) + e^-$	$\rightarrow Ag(s) + Br^-$	0.10
$2H^+ + 2e^-$	**$\rightarrow H_2(g)$**	**0.00**
$Pb^{2+} + 2e^-$	$\rightarrow Pb(s)$	-0.13
$Sn^{2+} + 2e^-$	$\rightarrow Sn(s)$	-0.14
$Ni^{2+} + 2e^-$	$\rightarrow Ni(s)$	-0.25
$Fe^{2+} + 2e^-$	$\rightarrow Fe(s)$	-0.44
$Cr^{3+} + 3e^-$	$\rightarrow Cr(s)$	-0.74
$Zn^{2+} + 2e^-$	$\rightarrow Zn(s)$	-0.76
$2H_2O + 2e^-$	$\rightarrow H_2(g) + 2OH^-$	-0.83
$Al^{3+} + 3e^-$	$\rightarrow Al(s)$	-1.66
$Mg^{2+} + 2e^-$	$\rightarrow Mg(s)$	-2.36
$Na^+ + e^-$	$\rightarrow Na(s)$	-2.71
$Ca^{2+} + 2e^-$	$\rightarrow Ca(s)$	-2.87
$K^+ + e^-$	$\rightarrow K(s)$	-2.93
$Li^+ + e^-$	$\rightarrow Li(s)$	-3.05

Increasing strength of oxidising agent ↑

Increasing strength of reducing agent ↓

This table is found helpful in the following facts.

(i) Predicting the Strength of Oxidant or Reductant

Since, the table is based on the decreasing order of standard electrode potential (i.e. reduction potential), i.e. the tendency of the substance to get reduced decreases on moving downwards. In other words, their reducing tendency increases on moving from top to bottom.

Thus, it is clear that an element can reduce all the elements placed above it in the series or it can be oxidised by all the elements present above it in the series. Thus,

- a negative $E°$ means that the redox couple is a stronger reducing agent than the H^+ / H_2 couple.
- a positive $E°$ means that the redox couple is a weaker reducing agent than the H^+ / H_2 couple.

EXAMPLE 13. Given the standard electrode potentials, $K^+ / K = -2.93V$, $Ag^+ / Ag = 0.80V$, $Hg^{2+} / Hg = 0.79V$, $Mg^{2+} / Mg = -2.37\ V$, $Cr^{3+} / Cr = -0.74\ V$
Arrange these metals in their increasing order of reducing power.

Note Lower the electrode potential ($E°_{red}$), stronger is the reducing agent.

Sol. The arrangement of metals in their increasing order of reducing power is as follows Ag < Hg < Cr < Mg < K.

(ii) Comparing the Reactivity of Metals

A metal having more negative value of electrode potential can displace the metal having less negative value of electrode potential from their salt solutions. e.g. Zn (with $E° -0.76$), can displace Cu (with $E° = +0.34$) from $CuSO_4$ solution but opposite is not true.

$$Zn(s) + CuSO_4(aq) \longrightarrow ZnSO_4(aq) + Cu(s)$$

Thus, zinc is more reactive than copper.

(iii) Predicting Feasibility of a Redox or Cell Reaction

The feasibility of a reaction is decided by its cell potential or emf ($E°_{cell} = E°_{cathode} - E°_{anode}$). A positive value of $E°_{cell}$ suggest that the cell reaction is feasible. However, if $E°_{cell}$ is negative or zero, the cell reaction will not exist.

EXAMPLE 14. Predict whether the following cell does exist or not?

$$Zn\,|\,Zn^{2+}(1M)\,||\,Cd^{2+}(1M)\,|\,Cd$$

[Given, $E°_{Zn,\,Zn^{2+}} = 0.76$ V and $E°_{Cd,\,Cd^{2+}} = 0.40$ V]

Sol. Follow the following steps to solve such problems.

Step I Obtain the value of $E°_{red}$ for both the redox couples (if not given).

Since, the given $E°$ values shows the oxidation electrode potential, so invert this value to get $E°_{red}$ (reduction electrode potential), i.e.

$$E°_{Zn^{2+}/Zn} = -0.76\ V;\ E°_{Cd^{2+}/Cd} = -0.40\ V$$

Step II Find cathode and anode for the cell.

Here, Zn / Zn^{2+} couple is anode and Cd^{2+} / Cd couple is cathode.

[If cell is not given, the couple having more negative $E°_{cell}$ or $E°$ value, acts as anode and the other one acts as cathode]

Step III Calculate the cell potential or emf using the expression, $E°_{red} = E°_{cathode} - E°_{anode}$

$$E°_{cell} = -0.40 - (-0.76)$$
$$= -0.40 + 0.76 = +0.36\ V$$

Step IV If $E°_{cell}$ is positive the reaction is feasible and the cell exists and if $E°_{cell}$ is negative or zero, the cell does not exist.

Since, we get positive $E°_{cell}$, so the cell will exist.

(iv) Predicting the Liberation of Hydrogen from an Acid

Highly reactive metals (i.e. metals having negative value of $E°$) displaces hydrogen from acids. Thus, metals like Mg, Al, Zn, Fe etc., replace hydrogen from HCl or H_2SO_4. Further, metals with positive value of $E°$ are not capable to replace hydrogen from mineral acids.

(v) Predicting the Product of Electrolysis

If electrolysis is carried out in aqueous solution, along with the cation, H^+ also tries to get reduced.

So, there is a competition between the metal ion and H^+ ions to get reduced.

Similarly, OH^- competes with the negative ion and electrode potential play an important role in deciding the product of electrolysis.

At anode, that species is oxidised, oxidation potential of which is higher or reduction potential of which is lower (more negative).

Similarly, at cathode that species is reduced, reduction potential of which is higher or oxidation potential of which is lower (i.e. less negative).

Note • In aqueous solution, having metal ions Li^+, Na^+, K^+, Ba^{2+}, Mg^{2+} or Al^{3+}, it is hydrogen ion that get reduced at cathode, thus, metal is not obtained.

- Similarly, if the solution contains SO_4^{2-} or NO_3^- ions, it is the water that get oxidised at anode,
  e.g. $2H_2O \longrightarrow O_2 + 4H^+ + 4e^-$

EXAMPLE 15. Predict the products of electrolysis in each of the following.

(i) An aqueous solution of $AgNO_3$ with silver electrodes.

(ii) An aqueous solution of $AgNO_3$ with platinum electrodes.

(iii) A dilute solution of H_2SO_4 with platinum electrodes.

(iv) An aqueous solution of $CuCl_2$ with platinum electrodes. **NCERT**

Hint:
- Higher the oxidation potential of an element, more readily it gets oxidised.
- In case of aqueous solution, H^+ and OH^- ions are also present, so compare the oxidation potential of the elements/ions with that of the H^+ and OH^- to decide the product.

Sol (i) **An aqueous solution of $AgNO_3$ with silver electrodes.**

Two oxidation and two reduction half reactions must be considered.

Oxidation (at anode)

(a) $Ag(s) \longrightarrow Ag^+(aq) + e^-$; $E° = -0.80$ V

(b) $2H_2O(l) \longrightarrow O_2(g) + 4H^+(aq) + 4e^-$; $E° = -1.23$ V

Reduction (at cathode)

(c) $Ag^+(aq) + e^- \longrightarrow Ag(s)$; $E° = +0.80$ V

(d) $2H_2O(l) + 2e^- \longrightarrow H_2(g) + 2OH^-(aq)$; $E° = -0.83$ V

By $E°$ values of (a) and (b), it appears that **at anode**, silver of silver anode gets oxidised more readily because oxidation potential of Ag is greater than that of H_2O molecule. Similarly, by $E°$ values of (c) and (d), it appears that **at cathode,** reduction potential of Ag^+ ions higher than that of H_2O molecules.

Therefore, on electrolysis of aqueous $AgNO_3$ solution with silver electrodes, Ag from silver anode dissolves while $Ag^+(aq)$ ions present in the solution gets reduced and deposited at cathode.

(ii) **An aqueous solution of $AgNO_3$ with platinum electrodes.** Platinum is an inert electrode so at anode oxidation of water takes place. As a result of this, O_2 is released at anode according to equation (b) [as in (i)]. At cathode, reduction of Ag^+ ions takes place.

Therefore, on electrolysis of aqueous $AgNO_3$ solution with platinum electrodes, O_2 is released at anode and Ag^+ ions from solution gets deposited at cathode.

(iii) **A dilute solution of H_2SO_4 with platinum electrodes.**

$H_2SO_4(aq) \rightleftharpoons 2H^+ + SO_4^{2-}$

When current is passed, either H^+ ions or H_2O molecules are reduced at cathode.

$2H^+(aq) + 2e^- \longrightarrow H_2(g)$; $E° = 0.0$ V

$2H_2O(aq) + 2e^- \longrightarrow H_2(g) + 2OH^-(aq)$; $E° = -0.83$ V

At cathode, H^+ ions gets reduced first because reduction potential of H^+ ions is greater than H_2O molecules. Similarly, on passing current either SO_4^{2-} ions or H_2O molecules are oxidised at anode. SO_4^{2-} ions are resistant to oxidation and are not discharged at anode.

So, at anode, H_2O molecules are oxidised to release O_2 gas.

$2H_2O(l) \longrightarrow O_2(g) + 4H^+(aq) + 4e^-$; $E° = -1.23$ V

Therefore, an electrolysis of an aqueous solution of H_2SO_4, only electrolysis of water occurs releasing H_2 at cathode and O_2 at anode.

(iv) **An aqueous solution of $CuCl_2$ with platinum electrodes.**

$CuCl_2(aq) \rightleftharpoons Cu^{2+} + 2Cl^-(aq)$;

(ionisation of $CuCl_2$ in water)

Two oxidation and two reduction half reactions must be considered.

Oxidation (at anode)

(a) $2Cl^-(aq) \longrightarrow Cl_2(g) + 2e^-$; $E° = -1.36$ V

(b) $2H_2O(l) \longrightarrow O_2(g) + 4H^+(aq) + 4e^-$; $E° = -1.23$ V

Reduction (at cathode)

(c) $Cu^{2+}(aq) + 2e^- \longrightarrow Cu(s)$; $E° = +0.34$ V

(d) $2H_2O(l) + 2e^- \longrightarrow H_2(g) + 2OH^-$; $E° = -0.83$ V

Although oxidation potential of H_2O molecule is greater than that of Cl^- ions, oxidation of Cl^- ions take place in preference to H_2O. It is because of the overvoltage of O_2.

Much lower potential than −1.36 V is required for oxidation of H_2O molecules.

Since, the reduction potential of Cu^{2+} is much higher than the reduction potential of H_2O molecules. therefore, Cu^{2+} ions are reduced at cathode.

Therefore, on electrolysis of an aqueous solution of $CuCl_2$, Cu metal is released at the cathode while Cl_2 gas is evolved at the anode.

Topic Practice 2

1. Write the oxidation and reduction half reactions from the following redox reaction.

$$2Fe + 2H_2O + O_2 \longrightarrow 2Fe(OH)_2$$

2. MnO_4^{2-} undergoes disproportionation reaction in acidic medium but MnO_4^- does not. Give reason. **NCERT**

3. How does Cu_2O act as both oxidant and reductant? Explain with proper reactions showing the change of oxidation numbers in each example.

4. The reaction

$$Cl_2(g) + 2OH^-(aq) \rightarrow ClO^-(aq) + Cl^-(aq) + H_2O(l)$$

represents the process of bleaching. Identify and name the species that bleaches the substances due to its oxidising action. **NCERT Exemplar**

Hint : Write the oxidation number of each element above its symbol and then identify the bleaching reagent by observing the change in oxidation number.

5. Identify the type of redox reaction taking place in the following.

(i) $3\overset{0}{Mg}(s) + \overset{0}{N_2}(g) \longrightarrow \overset{+2\,-3}{Mg_3N_2}(s)$

(ii) $\overset{+5\,-2}{V_2O_5}(s) + 5\overset{0}{Ca}(s) \longrightarrow 2\overset{0}{V}(s) + 5\overset{+2\,-2}{CaO}(s)$

(iii) $2\overset{+1\,+5\,-2}{KClO_3}(s) \longrightarrow 2\overset{+1\,-1}{KCl}(s) + 3\overset{0}{O_2}(g)$

(iv) $\overset{0}{Ca}(s) + 2\overset{+1\,-2}{H_2O}(l) \longrightarrow \overset{+2\,-2\,+1}{Ca(OH)_2}(aq) + \overset{0}{H_2}(g)$

(v) $\overset{0}{Br_2}(l) + 2\overset{-1}{I^-}(aq) \longrightarrow 2\overset{-1}{Br^-}(aq) + \overset{0}{I_2}(s)$

(vi) $\overset{0}{Cl_2}(g) + 2OH^-(aq) \longrightarrow \overset{-1}{ClO^-}(aq) + \overset{-1}{Cl}(aq) + H_2O(l)$

6. Find the value of n in

$$4\,MnO_4^- + 8H^+ + ne^- \longrightarrow Mn^{2+} + 4H_2O$$

7. Balance the following equations in basic medium by ion electron and oxidation number methods.

(i) $P_4(s) + OH^-(aq) \longrightarrow PH_3(g) + H_2PO_2^-(aq)$

(ii) $N_2H_4(l) + ClO_3^-(aq) \longrightarrow NO(g) + Cl^-(g)$

(iii) $Cl_2O_7(g) + H_2O_2(aq) \longrightarrow ClO_2(aq) + O_2(g)$

8. Write balanced chemical equation for the following reactions.

Permanganate ion (MnO_4^-) reacts with sulphur dioxide gas in acidic medium to produce Mn^{2+} and hydrogen sulphate ion. (Balance by ion electron method)

9. What would happen if no salt-bridge were used in the electrochemical cell (e.g. Zn -Cu cell)?

10. Can we use KCl as an electrolyte in the salt-bridge of the cell,

$$Cu(s) \mid Cu^{2+}(aq) \mid\mid Ag^+(aq) \mid Ag(s)$$

11. Write the anode reaction, the cathode reaction and the net cell reaction in the following cells. Which electrode would be positive terminal in each cell ?

(i) $Zn(s) \mid Zn^{2+} \mid\mid Br_2 \mid Br^- \mid Pt(s)$

(ii) $Cr(s) \mid Cr^{3+} \mid\mid I_2 \mid I^- \mid Pt(s)$

(iii) $Pt(s) \mid H_2(g) \mid H^+(aq) \mid\mid Cu^{2+} \mid Cu(s)$

12. Following cell is set up between copper and silver electrodes.

$$Cu \mid Cu^{2+}(aq) \mid\mid Ag^+(aq) \mid Ag$$

If its two half-cells work under standard conditions, calculate the emf of the cell.

[Given: $E^\circ_{Cu^{2+}/Cu} = +0.34$ V, $E^\circ_{Ag^+/Ag} = +0.80$ V]

13. Represent a galvanic cell in electrode and ions containing Cu electrode dipped in molar solution of copper sulphate and silver electrode dipped in molar solution of silver nitrate.

[Given: $E^\circ_{Cu^{2+}/Cu(s)} = 0.34$ V, $E^\circ_{Ag^+/Ag(s)} = 0.80$ V]

14. Out of aluminium and silver vessel, which one will be more suitable to store 1 M HCl solution and why ?

$$E^\circ_{Al^{3+}|Al} = -1.66\text{ V}, E^\circ_{Ag^+|Ag} = +0.80\text{ V}$$

15. Can Fe^{3+} oxidise Br^- to Br_2 at 1 M concentrations?

$E^\circ_{(Fe^{3+}/Fe^{2+})} = 0.77$ V and $E^\circ_{(Br/Br^-)} = 1.09$ V

16. How would you know whether a redox reaction is taking place in an acidic, alkaline or neutral medium?

17. Arrange the following metals in the order in which they displace each other from the solution of their salts.

Al, Cu, Fe, Mg and Zn **NCERT**

18. Two half-cells are $Al^{3+}(aq)$ / Al and $Mg^{2+}(aq)$ / Mg. The reduction potentials of these half-cells are – 1.66 V and – 2.36 V respectively. Calculate the cell potential. Write the cell reaction also.

19. What sorts of informations can you draw from the following reaction? **NCERT**

$$(CN)_2(g) + 2OH^-(aq) \longrightarrow CN^-(aq) + CNO^-(aq) + H_2O(l)$$

20. Chlorine is used to purify drinking water. Excess of chlorine is harmful. The excess of chlorine is removed by treating with sulphur dioxide. Present a balanced equation for this redox change taking place in water. **NCERT**

21. How many millimoles of potassium dichromate is required to oxidise 24 mL of 0.5 M Mohr's salt solution in acidic medium?

22. In Ostwald's process for the manufacture of nitric acid, the first step involves the oxidation of ammonia gas by oxygen gas to give nitric oxide gas and steam. What is the maximum weight of nitric oxide that can be obtained starting only with 10.00 g of ammonia and 20.00 g of oxygen? **NCERT**

Hint:
- The amount of NO produced is decided by the amount of limiting reagent. So first find limiting reagent by writin3g a balanced chemical reaction and applying unitary method.
- Then, find the amount of NO produced from the limiting reagent.

23. Justify giving reactions that among halogens, fluorine is the best oxidant and among hydrohalic compounds, hydroiodic acid is the best reductant. **NCERT**

Hint:
- Higher is the standard electrode potential, higher is the oxidising power.
- In case of halogens, standard reduction potential decreases from fluorine to iodine while in case of halide ions, the order is reverse.

24. Copper dissolves in dilute nitric acid but not in dilute HCl. Explain.

Summary

- The reactions involving oxidation and reduction as their two half-reactions is known as **redox reaction**.
- **Oxidation** involves addition of oxygen or other electronegative element, removal of H or other electropositive element, loss of electron and increase in oxidation number.
- **Reduction** is just opposite to that of oxidation, i.e. involves addition of H or other electropositive element, gain of electron etc.
- **Oxidising agent** is the species, which undergoes reduction.
- **Reducing agent** undergoes oxidation.
- **Oxidation number** (O.N) is the charge that an atom appear to have in a compound when all other atoms are removed.
  - (i) It is zero for free elements (i.e. as exist in nature). e.g. P_4, O_2, S_8 etc.
  - (ii) It is equal to group number for elements of group 1 and 2.
  - (iii) Algebric sum of oxidation numbers = 0 (for compounds)= charge (for ions).
- **Paradox of fractional oxidation number** In C_3O_2, O.N. of two C-atoms is +2 and one C-atom is zero(0).
- Stock notation = cation + (oxidation number in roman numerals) + anion. e.g. Au(I)Cl.
- **Types of redox reactions**
  - (i) **Combination reactions** involve combination of two or more atoms to give a single product.
  - (ii) **Decomposition reactions** These involve splitting of a compound into two or more simpler products. However, all such reactions are not the decomposition reactions.
  - (iii) **Displacement reactions** involve displacement of one atom (or ion) by other.
    - I. **Metal displacement reactions** Here, one metal displaces the other metal.
    - II. **Non-metal displacement reactions** Here, non-metal is displaced by metal or non-metal.
  - (iv) **Disproportionation reactions** In these reactions the same element is oxidised as well as reduced.
- **Balancing of redox reactions**
  - (i) **Oxidation number method** First write ionic equation then assign oxidation number. Balance all other atoms except H and O, then predict change in oxidation number balance charge balance H and O.
  - (ii) **Ion electron method** First write ionic equation and split it into two halves. Balance all other atoms except H and O, (in both halves) balance charge (in both halves) and add both half equations verify balanced equation and convert ionic equation into molecular form.
- The titration involving redox reactions are called **redox-titrations**.
- **Electrolytic cell** converts electrical energy into chemical energy.
- **Electrochemical cell** converts chemical energy into electrical energy. e.g. Daniell cell, lead storage battery etc.
- **Salt-bridge** completes the circuit and maintain electrical neutrality.
- **Some terminology**
  - (i) A metal rod dipped in its salt solution is called **electrode**.
  - (ii) The electrode at which oxidation occurs is called **anode**.
  - (iii) The electrode at which reduction occurs is called **cathode**.
  - (iv) **Redox couple** is the combination of oxidised and reduced forms. e.g. Zn^{2+}/Zn.
  - (v) **Electrode potential** is the potential associated with an electrode.
- **Standard electrode potential** is the electrode potential measures in standard conditions.
- The arrangement of various elements in order of increasing values of standard reduction potentials is called electrochemical **series or activity series**.

Exam Pattern Exercise

Part A - Objective Questions

Multiple Choice Questions

1. From the given species such as Li, K, Ca and Na, which of the following is the strongest reducing agent?
(a) Na (b) Li (c) Ca (d) K

2. Which is the best description of behaviour of bromine in the given equation?

$$H_2O + Br_2 \longrightarrow HBr + HOBr$$

(a) Proton acceptor
(b) Both oxidised and reduced
(c) Only oxidised
(d) Only reduced

3. In the reaction,
$2KClO_3 \longrightarrow 2KCl + 3O_2$, the elements which have been oxidised and reduced respectively are
(a) chlorine and oxygen
(b) oxygen and chlorine
(c) potassium and oxygen
(d) oxygen and potassium

4. The largest oxidation number exhibited by an element depends on its outer electronic configuration. With which of the following outer electronic configurations the element will exhibit largest oxidation number?
NCERT Exemplar
(a) $3d^1 4s^2$ (b) $3d^3 4s^2$
(c) $3d^5 4s^1$ (d) $3d^5 4s^2$

5. Water molecule is formed by the reaction,

$$2H_2 + O_2 \longrightarrow 2H_2O,$$

What does happen in this reaction?
(a) Electrons are transferred from H to O-atom
(b) Electrons are transferred from O to H-atom
(c) Electrons are accepted by H from O-atom
(d) Electrons are donated by O to H-atom

6. In which of the following oxidation number of chlorine is +5 ?
(a) Cl^- (b) ClO^- (c) ClO_2^- (d) ClO_3^-

7. Which of the following species contain an element in an oxidation state that is not a whole number?
(a) VO_4^{3-} (b) Mn_2O_3 (c) $S_4O_6^{2-}$ (d) Cl_2O_7

8. In the redox reaction,

$$MnO_4^- + 8H^+ + 5Br^- \longrightarrow Mn^{2+} + 4H_2O + \frac{5}{2}Br_2$$

Which one is the reducing agent?
(a) H^+ (b) MnO_4^-
(c) Br^- (d) Mn^{2+}

9. The reaction, $2H_2O(l) \xrightarrow{\Delta} 2H_2(g) + O_2(g)$ is an example of
(a) addition reaction (b) decomposition reaction
(c) displacement reaction (d) None of these

10. Identify disproportionation reaction
NCERT Exemplar
(a) $CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O$
(b) $CH_4 + 4Cl_2 \longrightarrow CCl_4 + 4HCl$
(c) $2F_2 + 2OH^- \longrightarrow 2F^- + OF_2 + H_2O$
(d) $2NO_2 + 2OH^- \longrightarrow NO_2^- + NO_3^- + H_2O$

11. In the reaction, $8Al + 3Fe_3O_4 \longrightarrow 4Al_2O_3 + 9Fe$ the number of electrons transferred from the reductant to the oxidant is
(a) 8 (b) 4
(c) 16 (d) 24

12. The value of 'n' in the reaction,
$Cr_2O_7^{2-} + 14H^+ + nFe^{2+} \longrightarrow 2Cr^{3+} + nFe^{3+} + 7H_2O$ will be
(a) 2 (b) 3 (c) 6 (d) 7

13. Solution of potassium chloride or ammonium nitrate in salt-bridge usually solidified by boiling with
(a) *agar-agar* (b) starch
(c) cellulose (d) glycogen

14. Using the standard electrode potential, find out the pair between which redox reaction is not feasible.
$E^\ominus$ values: $Fe^{3+}/Fe^{2+} = +0.77$; $I_2/I^- = +0.54$; $Cu^{2+}/Cu = +0.34$; $Ag^+/Ag = +0.80$ V
NCERT Exemplar
(a) Fe^{3+} and I^- (b) Ag^+ and Cu
(c) Fe^{3+} and Cu (d) Ag and Fe^{3+}

15. Negative $E^\ominus$ indicates that redox couple is
(a) weaker reducing agent than H^+/H_2 couple
(b) stronger reducing agent than H^+/H_2 couple
(c) stronger oxidising agent than H^+/H_2 couple
(d) None of the above

16. Consider the given standard electrode potentials.

I. $K^+/K = -3.02$ V

II. $Cu^{2+}/Cu = +0.34$ V

III. $Hg^{2+}/Hg = 0.92$ V

IV. $Cr^{3+}/Cr = -0.74$ V

Decreasing order of reducing power of these elements is

(a) I > II > III > IV (b) I > IV > III > II

(c) I > IV > II > III (d) III > II > IV > I

Assertion-Reason

■ **Directions** (Q. Nos. 17-22) In the following questions. an Assertion (A) is followed by a corresponding Reason (R) Use the following keys to choose the appropriate answer.

(a) Both (A) and (R) are correct and (R) is the correct explanation of (A).

(b) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

(c) (A) is correct and (R) is incorrect.

(d) (A) is incorrect and (R) is correct.

17. **Assertion** (A) Lithium ion is a poor oxidising agent.

Reason (R) Lithium ion has the lowest electrode potential.

18. **Assertion** (A) Bromide ion acts as a reducing agent in given following reaction.

$$2MnO_4^-(aq) + Br^-(aq) + H_2O(l) \longrightarrow MnO_2(aq) + BrO_3^-(aq) + 2OH^-(aq)$$

Reason (R) Oxidation number of Br increases from –1 to +5 .

19. **Assertion** (A) In the representation $E^{\ominus}_{Fe^{3+}/Fe^{2+}}$ and $E^{\ominus}_{Cu^{3+}/Cu}$ are redox couples.

Reason (R) Redox couple is the combination of oxidised and reduced form of a substance involved in an oxidation or reduction half-cell.

20. **Assertion** (A) Among halogens, fluorine is the best oxidant.

Reason (R) Fluorine is the most electronegative atom.

21. **Assertion** (A) Fe reacts with HCl to produce H_2 gas.

Reason (R) Fe is a better oxidising agent thant H_2.

22. **Assertion** (A) When Zn rod is dipped in $Cu(NO_3)_2$ solution, blue colour of solution starts fading.

Reason (R) In this reaction, Zn gets reduced and Cu^{2+} ions gets oxidised.

Case-Study

23. The concept of electron transfer is found unable to explain the redox changes or electron shift in case of covalent compounds. To explain these changes a new concept, called **oxidation number** is introduced. Oxidation number is defined as the charge that an atom of the element has in its ion or appear to have when present in the combined state with other atoms.

Each atom that participates in an oxidation-reduction is assigned an oxidation number that reflects its ability to acquire, donate or share electrons. Certain elements assume the same oxidation number in different compounds. Other notably the non-metals and the transition elements, can assume a variety of oxidation numbers. In the nomenclature of inorganic chemistry, the oxidation number of an element that may exist is more than one oxidation state is indicated by a roman numeral in parenthesis after the name of the element.

(i) The oxidation number of underlined element in $K\underline{O}_2$ is

(a) 0 (b) 2 (c) –1 (d) –1/2

(ii) The element which never has positive oxidation number in any of its compound is

(a) F (b) Cl (c) O (d) N

(iii) The brown ring complex compound is formulated as $[Fe(H_2O)_5NO]SO_4$. The oxidation state of iron in the given complex is

(a) +1 (b) –3 (c) +2 (d) –2

Or When $KMnO_4$ is reduced with oxalic acid in acidic medium, the oxidation number of Mn change from

(a) +7 to +2 (b) +7 to +4

(c) +4 to +2 (d) +7 to +5

(iv) The oxidation state of sulphur in sodium tetrathionate $[Na_2S_4O_6]$ is

(a) 2 (b) 0 (c) 2.5 (d) 3.5

24. In a redox reaction, the substance which oxidises the other or which itself undergoes reduction is called the oxidising agent and the substance that reduces the other and itself undergoes oxidation is called the reducing agent.

e.g. $2HgCl_2 + SnCl_2 \longrightarrow Hg_2Cl_2 + SnCl_4$

Thus, in the above example, $HgCl_2$ is an oxidising agent (as it reduces) and $SnCl_2$ is a reducing agent (as it oxidises).

In these questions (i)-(iv) a statement of Assertion (A) followed by a statement of Reason (R) is given. Choose the correct answer out of the following choices :

(a) Both (A) and (R) are correct and (R) is the correct explanation of (A).

(b) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

(c) (A) is correct and (R) is incorrect.

(d) (A) is incorrect and (R) is correct.

(i) **Assertion** (A) This reaction is redox reaction,

$$K_2Cr_2O_7(aq) + 3SO_2(g) + H_2SO_4(aq) \longrightarrow Cr_2(SO_4)_3(aq) + K_2SO_4(aq) + H_2O(l)$$

Reason (R) Both oxidation and reduction take place in the given reaction.

(ii) **Assertion** (A) The decomposition of hydrogen peroxide to form water and oxygen is an example of disproportionation reaction.

Reason (R) The oxygen of peroxide is in –1 oxidation state and it is converted to zero oxidation state in O_2 and –2 oxidation state in H_2O.

(iii) **Assertion** (A) In the species, Br_3O_8 each of two extreme bromine exhibits oxidation state of +6 and the middle bromine of +4.

Reason (R) The average of three oxidation numbers of bromine of the Br_3O_8 is 16/3.

(iv) **Assertion** (A) In the reaction between potassium permanganate and potassium iodide, permanganate ions act as oxidising agent.

Reason (R) Oxidation state of manganese changes from +2 to +7 during the reaction.

Or **Assertion** (A) Zinc is used to reduce iron.

Reason (R) The E_{oxi} value of Zn is less than E_{oxi} of Fe.

Part B - Subjective Questions

Short Answer Type Questions

1. Calculate the oxidation number of phosphorus in the following species.

   (i) HPO_3^{2-} (ii) PO_4^{3-} **NCERT Exemplar**

2. Why does fluorine not show disproportionation reaction? **NCERT Exemplar**

3. Fluorine reacts with ice and results in the change.

   $$H_2O(s) + F_2(g) \longrightarrow HF(g) + HOF(g)$$

   Justify that this reaction is a redox reaction. **NCERT**

   Hint: Redox reactions involve oxidation and reduction as its two half reactions. So, find the oxidation state of each element given in the equation to find which substance is oxidising and which is reducing, then check, if the given reaction is redox reaction or not.

4. Nitric acid is an oxidising agent and reacts with PbO but it does not react with PbO_2. Explain why? **NCERT**

5. Calculate the oxidation number of sulphur, chromium and nitrogen in H_2SO_5, $Cr_2O_7^{2-}$ and NO_3^-. Suggest the structure of these compounds. Count for the fallacy. **NCERT**

6. Write formulae for the following compounds.

   (i) Mercury (II) chloride
   (ii) Nickel (II) sulphate
   (iii) Tin (IV) oxide
   (iv) Thallium (I) sulphate
   (v) Iron (III) sulphate
   (vi) Chromium (III) oxide **NCERT**

   Hint:
   - While writing the formula of a compound, oxidation state (or valency) of cation is written as the subscript of anion and valency of anion is written as the subscript of cation.
   - Valency of chloride (Cl^-), sulphate (SO_4^{2-}) and oxide (O^{2-}) are – 1, – 2 and – 2 respectively. Valencies of metals are given in brackets.

7. While sulphur dioxide and hydrogen peroxide can act as oxidising as well as reducing agents in their reactions, ozone and nitric acid act only as oxidants. Why? **NCERT**

   Hint : An element, in its lowest oxidation state, can behave only as reductant while in its highest oxidation state can behave only as oxidants. In intermediate oxidation states, elements can behave as oxidant as well as reductant.

8. What are the net charges on the left and right side of the following equations? Add electrons as necessary to make each of them balanced half reactions.

   (i) $NO_3^- + 10H^+ \longrightarrow NH_4^+ + 3H_2O$

   (ii) $Cl_2 + 4H_2O \longrightarrow 2ClO_2^- + 8H^+$

9. The Mn^{3+} ion is unstable in solution and undergoes disproportionation to give Mn^{2+}, MnO_2 and H^+ ions. Write a balanced ionic equation for the reaction.

10. A solution of silver nitrate was stirred with iron rod. Will it cause any change in the concentration of silver and nitrate ions?

11. An iron rod is immersed in solution containing 1.0 M $NiSO_4$ and 1.0 M $ZnSO_4$. Predict giving reasons which of the following reactions is likely to proceed?
(i) Fe reduces Zn^{2+} ions,
(ii) Iron reduces Ni^{2+} ions.
Given, $E^\circ_{Zn^{2+}/Zn} = -0.76$ V, $E^\circ_{Fe^{2+}/Fe} = -0.44$ V
and $E^\circ_{Ni^{2+}/Ni} = -0.25$ V

12. Which method can be used to find out strength of reductant/oxidant in a solution ? Explain with an example. **NCERT Exemplar**

13. Depict the galvanic cell in which the reaction,
$Zn(s) + 2Ag^+(aq) \longrightarrow Zn^{2+}(aq) + 2Ag(s)$
takes place, further show **NCERT**
(i) which of the electrode is negatively charged?
(ii) the carriers of the current in the cell.
(iii) individual reaction at each electrode.

Long Answer Type Questions

14. What do you understand by oxidation and reduction? Classify the following substances into oxidising and reducing agents.
(i) Carbon (ii) Ozone
(iii) Nascent hydrogen (iv) Nitric acid
(v) Chlorine

15. Using electron transfer concept, identify the oxidant and reductant in the following redox reactions.
(i) $Zn(s) + 2H^+(aq) \longrightarrow Zn^{2+}(aq) + H_2(g)$
(ii) $2[Fe(CN)_6]^{4-}(aq) + H_2O_2(aq) + 2H^+(aq) \longrightarrow 2[Fe(CN)_6]^{3-}(aq) + 2H_2O\ (l)$
(iii) $2[Fe(CN)_6]^{3-}(aq) + 2OH^-(aq) + H_2O_2(aq) \longrightarrow 2[Fe(CN)_6]^{4-}(aq) + O_2(g) + 2H_2O\ (l)$
(iv) $BrO_3^-(aq) + F_2(g) + 2OH^-(aq) \longrightarrow BrO_4^-(aq) + 2F^-(aq) + H_2O(l)$
(v) $2NaClO_3(aq) + I_2(aq) \longrightarrow 2NaIO_3(aq) + Cl_2(g)$

16. What are the oxidation numbers of the underlined elements in each of the following and how do you rationalise your results?
(i) $K\underline{I}_3$
(ii) $H_2\underline{S}_4O_6$
(iii) $\underline{Fe}_3O_4$
(iv) $\underline{C}H_3\underline{C}H_2OH$
(v) $\underline{C}H_3\underline{C}\ OOH$ **NCERT**

17. Justify that the following reactions are redox reactions. **NCERT**
(i) $CuO(s) + H_2(g) \longrightarrow Cu(s) + H_2O(g)$
(ii) $Fe_2O_3(s) + 3CO(g) \longrightarrow 2Fe(s) + 3CO_2(g)$
(iii) $4BCl_3(g) + 3LiAlH_4(s) \longrightarrow 2B_2H_6(g) + 3LiCl(s) + 3AlCl_3(s)$
(iv) $2K(s) + F_2(g) \longrightarrow 2K^+F^-(s)$
(v) $4NH_3(g) + 5O_2(g) \longrightarrow 4NO(g) + 6H_2O(g)$

18. Write correctly the balanced equations for the following redox reactions using half reactions.
(i) $H_2S + Fe^{3+} \longrightarrow Fe^{2+} + S + H^+$
(ii) $I^- + IO_3^- + H^+ \longrightarrow I_2 + H_2O$
(iii) $Bi(s) + NO_3^- + H^+ \longrightarrow NO_2 + Bi^{3+} + H_2O$
(iv) $I^- + O_2(g) + H_2O \longrightarrow I_2 + OH^-$
State what is oxidised to and what is reduced in the reactions expressed by the equations?

19. Balance the following redox reactions by ion-electron method.
(i) $MnO_4^-(aq) + I^-(aq) \longrightarrow MnO_2(s) + I_2(s)$ (in basic medium)
(ii) $MnO_4^-(aq) + SO_2(g) \longrightarrow Mn^{2+}(aq) + HSO_4^-(aq)$ (in acidic solution)
(iii) $H_2O_2(aq) + Fe^{2+}(aq) \longrightarrow Fe^{3+}(aq) + H_2O(l)$ (in acidic solution)
(iv) $Cr_2O_7^{2-}(aq) + SO_2(g) \longrightarrow Cr^{3+}(aq) + SO_4^{2-}(aq)$ (in acidic solution)
NCERT

20. How do you account for the following observations?
(i) Though alkaline potassium permanganate and acidic potassium permanganate both are used as oxidants, yet in the manufacture of benzoic acid from toluene we use alcoholic potassium permanganate as an oxidant. Why? Write a balance redox equation for the reaction.
(ii) When concentrated sulphuric acid is added to an inorganic mixture containing chloride, we get colourless pungent smelling gas HCl, but if the mixture contains bromide, then we get red vapour of bromine. Why? **NCERT**

Hint:
- Like dissolves like, i.e. benzoic acid and toluene being covalent readily dissolve in alcohol and form homogeneous solution.
- HBr is a stronger reducing agent then HCl, so reduces H_2SO_4 to SO_2.

21. (i) Use the following reactions to arrange the elements A, B, C and D in order of their redox reactivity.

(a) $A + B^+ \longrightarrow A^+ + B$

(b) $B + D^+ \longrightarrow B^+ + D$

(c) $C^+ + D \longrightarrow$ No reaction

(d) $B + C^+ \longrightarrow B^+ + C$

(ii) On the basis of above redox activity series, predict which of the following reactions would you expect to occur?

(a) $A^+ + C \longrightarrow A + C^+$

(b) $A^+ + D \longrightarrow A + D^+$

Case-Study

22. The oxidation state of an individual atom in its free state is 0. The total oxidation state of all atoms in a neutral species is 0 and in an ions is equal to the charge. Group 1 metals have an oxidation state of +1 and group 2 an oxidation state of +2.

The oxidation state of fluorine is –1 in compounds. Hydrogen generally has an oxidation state of +1 in compounds.

Oxygen generally has an oxidation state of –2 in compounds.

In binary metal compounds, group 17 elements have an oxidation state of –1, group 16 elements of –2, and group 15 element of –3. The sum of the oxidation states is equal to zero for neutral compounds and equal to the charge for polyatomic ion species.

(i) How many moles of I_2 is liberated when one mole of acidified $K_2Cr_2O_7$ is reacted with excess KI?

(ii) What is an oxidising agent? Explain with the help of an equation.

(iii) How many moles of H_2O_2 are oxidised by mole of ClO_2 when ClO_2 oxidises H_2O_2 to O_2 in alkaline medium?

(iv) What is the change occur in oxidation state during reduction and oxidation ?

Or

What is the oxidation state of N in HNO_2?

23. The concept of the electron transfer is found unable to explain the redox changes or electron shift in case of covalent compounds.

To explain these changes a new concept, called oxidation number is introduced.

The oxidation state is the simplest attribute of an element in a compound. It is taught early in the chemistry curriculum as convenient electron-counting scheme for redox reactions. Its applications range from descriptive chemistry of elements to nomenclature and electrochemistry, or as an independent variable in plots and databases of bonded-atom properties.

The history of the oxidation state goes back about 200 years when it described the stepwise increase in the amount of oxygen bound by elements that form more than one oxide. In his 1835 texbook *Unorganische chemie*, Wöheler speaks of such an "oxydationsstufe". This expression remains in use for oxidation state in several languages. The equivalent term oxidation number is also common; in English this refers more to redox balancing than to the chemical systematics of an element.

(i) Name the compund of Mn in which it shows highest oxidation state.

(ii) Name the element which generally show negative oxidation state?

(iii) What is the oxidation state of Fe in brown ring complex?

(iv) What is the change in oxidation number of Mn when manganous salt is fused with a mixture of KNO_3 and solid NaOH?

Or

What is the oxidation state of I in IPO_4?

Hints and Solutions

Topic Practice 1

1. In the reaction, Fe_2O_3 loses oxygen and is reduced to Fe.

2. In AgF_2, Ag is in + 2 oxidation state. It is highly unstable so it readily accepts an electron to attain + 1 oxidation state which is more stable.

$$Ag^{2+} + e^- \longrightarrow Ag^+$$

That's why AgF_2 acts as a strong oxidising agent.

3. Since, oxygen is being added to Cu, therefore Cu_2S is oxidised to Cu_2O and the other reactant, i.e. O_2 is getting reduced.

Hence, Cu_2S is a reducing agent and O_2 is an oxidising agent.

4. (i) **HNO_3**

Oxidation number of N in HNO_3 is + 5.

Maximum oxidation number of N is + 5 because it has five electrons in the valence shell ($2s^2 2p^3$).

Minimum oxidation number of N is − 3 because it can accept 3 more electrons to get noble gas configuration.

Since, oxidation number of N in HNO_3 is maximum, therefore, it can only decrease. Thus, HNO_3 can act as an oxidising agent.

(ii) **HNO_2**

Oxidation number of N = + 3

Maximum oxidation number of N = + 5

Minimum oxidation number of N = − 3

Therefore, the oxidation number of N can increase by losing electrons or can decrease by accepting electrons. Thus, HNO_2 can act both as an oxidising as well as a reducing agent.

5. $\overset{+8}{Xe}O_6^{4-}(aq) + 2\overset{-1}{F}^-(aq) + 6H^+(aq) \longrightarrow$

$$\overset{+6}{Xe}O_3(g) + \overset{0}{F_2} + 3H_2O(l)$$

In the above reaction, oxidation number of Xe in XeO_6^{4-} decreases from + 8 to + 6 in XeO_3 and oxidation number of F increases from − 1 (in F^-) to zero (in F_2).

Hence XeO_6^{4-} or Na_4XeO_6 is reduced and F^- is oxidised. This reaction occur because Na_4XeO_6 or XeO_6^{4-} is a stronger oxidising agent than fluorine.

6. $2\overset{+2\,-2}{S_2O_3^{2-}}(aq) + \overset{0}{I_2}(s) \longrightarrow \overset{2.5\,-2}{S_4O_6^{2-}}(aq) + 2I^-(aq)$

$\overset{+2\,-2}{S_2O_3^{2-}}(aq) + 2\overset{0}{Br_2}(l) + 5H_2O(l) \longrightarrow$

$$2\overset{+6\,-2}{SO_4^{2-}}(aq) + 4Br^-(aq) + 10H^+(aq)$$

Bromine is a stronger oxidising agent in comparison to I_2. It oxidises S of $S_2O_3^{2-}$ to a higher oxidation state + 6 in SO_4^{2-}. While I_2 oxidises S of $S_2O_3^{2-}$ to oxidation state 2.5 in $S_4O_6^{2-}$. That's why same reductant, thiosulphate react differently with bromine and iodine.

7. (i) P_4 is a reducing agent and Cl_2 is an oxidising agent.

(a) $\underset{\text{Excess}}{\overset{0}{P_4}}(s) + 6Cl_2(g) \longrightarrow \underset{\substack{\text{Lower oxidation}\\ \text{state of P}}}{\overset{+3}{4PCl_3}}$

(b) $\overset{0}{P_4}(s) + \underset{\text{Excess}}{10Cl_2} \longrightarrow \underset{\substack{\text{Higher oxidation}\\ \text{state of P}}}{\overset{+5}{4PCl_5}}$

Therefore, when P_4 (reducing agent) is in excess, PCl_3 is formed in which oxidation state of P is + 3 and if Cl_2 (oxidising agent) is in excess, PCl_5 is formed in which oxidation state of P is + 5. Other two examples are

(ii) C is a reducing agent, while O_2 is an oxidising agent.

(a) $\underset{\text{Excess}}{\overset{0}{2C}}(s) + O_2(g) \longrightarrow \overset{+2}{2CO}(g)$

(b) $\underset{\text{Excess}}{\overset{0}{C}}(s) + O_2 \longrightarrow \overset{+4}{CO_2}(g)$

When reducing agent C is in excess, a compound CO of lower oxidation state is formed, if oxidising agent O_2 is in excess, a compound CO_2 of higher oxidation state is formed.

(iii) Na is a reducing agent while O_2 is an oxidising agent.

(a) $\underset{\text{Excess}}{4Na}(s) + \overset{0}{O_2}(g) \longrightarrow \underset{\text{(lower oxidation state)}}{\overset{-2}{Na_2O}(s)}$

(b) $2Na(s) + \underset{\text{Excess}}{2O_2}(g) \longrightarrow \underset{\text{(higher oxidation state)}}{\overset{-1}{Na_2O_2}(s)}$

8. No, the oxidation number of an element in any species is an apparent charge on the atom which it appears to have acquired when all other atoms in the species are removed as ions.

9. (i) NaH_2PO_4

Let the oxidation number of P be x. Writing the oxidation number of each atom above its symbol, we get $\overset{+1}{Na}\overset{+1}{H_2}\overset{x}{P}\overset{-2}{O_4}$.

In neutral compounds, the sum of the oxidation numbers of all the atoms is zero.

$$1(+1) + 2(+1) + x + 4(-2) = 0$$
$$3 + x + (-8) = 0$$
$$x = 8 - 3 = +5$$

Hence, the oxidation number of P in NaH_2PO_4 is + 5.

Calculate oxidation number of other elements in the same way as shown in (i).

You will get oxidation number of S, P, Mn, O, B, S, S Cr, S in the given compound as +6, + 5, + 6, −1, + 3, + 6, + 6, + 6, + 2 respectively.

Note H_2O is a neutral molecule, therefore sum of oxidation numbers of all atoms in H_2O is zero.

Hence, the oxidation number of S in $KAl(SO_4)_2 \cdot 12H_2O$ is +6.

16. (i) F exhibits only negative oxidation state because it is the most electronegative element.

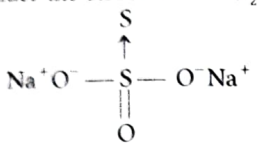
(ii) Cs exhibits only positive oxidation state because it is the most electropositive element.

(iii) I exhibits both positive and negative oxidation states. Iodine exhibits – 1, 0, + 1, 3, + 5 and + 7 oxidation states (+ 3, + 5 and + 7 oxidation states are exhibited by I due to the presence of vacant *d*-orbitals).

(iv) Ne is an inert gas, so it neither exhibits negative nor positive oxidation states.

17. (i) **$Na_2S_2O_3$**

Let us consider the structure of $Na_2S_2O_3$.



There is a coordinate bond between two sulphur atoms. The oxidation number of acceptor S-atom is – 2. Let the oxidation number of other S-atom be *x*.

$$\underset{\text{For Na}}{2(+1)} + \underset{\text{For O-atoms}}{3\times(-2)} + x + \underset{\substack{\text{For coordinate}\\ \text{S-atom}}}{1(-2)} = 0$$

$$x = +6$$

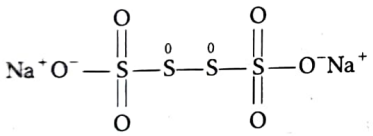
Therefore, the two sulphur atoms in $Na_2S_2O_3$ have – 2 and + 6 oxidation number.

The average oxidation number of two S-atom

$$= \frac{-2+6}{2} = +2$$

(ii) **$Na_2S_4O_6$**

Let us consider the structure of $Na_2S_4O_6$.



In this structure, two central sulphur atoms have zero oxidation number because electron pair forming the S—S bond remain in the centre. Let the oxidation number of (remaining S-atoms) S-atom be *x*.

$$\underset{\text{For Na}}{2(+1)} + \underset{\text{For O}}{6(-2)} + 2x + \underset{\substack{\text{For two}\\ \text{central}\\ \text{S-atom}}}{2(0)} = 0$$

$$2 - 12 + 2x = 0 \text{ or } x = +\frac{10}{2} = +5$$

Therefore, the two central S-atoms have zero oxidation state and two terminal S-atoms have + 5 oxidation state each.

The average oxidation number of four S-atom

$$= \frac{5+5+0+0}{4} = \frac{5}{2}$$

(iii) **Na_2SO_3**

Let the oxidation number of S in Na_2SO_3 be *x*.

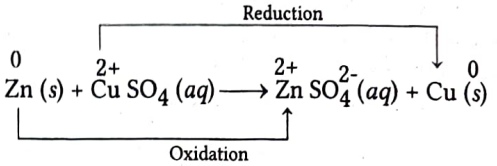
$$2(+1) + x + 3(-2) = 0 \text{ or } x = +4$$

(iv) **Na_2SO_4**

Let the oxidation number of S be *x*.

$$2(+1) + x + 4(-2) = 0 \text{ or } x = +6$$

12. The blue colour of $CuSO_4$ solution will get discharged and reddish brown copper metal will be deposited on Zn rod. This is because blue colour Cu^{2+} (in $CuSO_4$) gets reduced to Cu by accepting two electrons from Zn, which gets oxidised to colourless $ZnSO_4$.



13. $1\times(+3) + 2\times(+2) + 3x + 7\times(-2) = 0$

or $3 + 4 + 3x - 14 = 0$

or $3x = 7;\ x = 7/3$

14. In reactions, (i) and (ii), $AgNO_3$ and $CuSO_4$ act as oxidising agents respectively.

They oxidise H_3PO_2 (hypophosphorus acid) to (orthophosphoric acid). In reaction (iii), $[Ag(NH_3)_2]^+(aq)$ oxidises benzaldehyde to benzoic acid but in reaction (iv), Cu^{2+} do not oxidise benzaldehyde (C_6H_5CHO) to benzoic acid.

This indicates that Ag^+ is a stronger oxidising agent than Cu^{2+}.

15. Pb_3O_4 is actually a stoichiometric mixture of 2 moles of PbO and 1 mole of PbO_2, i.e. ($2\,PbO\cdot PbO_2$). In PbO_2, lead is present in + 4 oxidation state, whereas the stable oxidation state of lead in PbO is + 2. PbO_2 thus can act as an oxidant (oxidising agent) and therefore, can oxidise Cl^- ion of HCl into chlorine.

We may also keep in mind that PbO is a basic oxide. Therefore, the reaction

$$Pb_3O_4 + 8HCl \longrightarrow 3PbCl_2 + Cl_2 + 4H_2O$$

can be splitted into two reactions namely

$$2\,PbO + 4\,HCl \longrightarrow 2\,PbCl_2 + 2H_2O$$

(acid-base reaction)

$$\overset{+4}{PbO_2} + 4\overset{-1}{HCl} \longrightarrow \overset{+2}{Pb}\,Cl_2 + \overset{0}{Cl_2} + 2H_2O$$

(redox reaction)

Since, HNO_3 itself is an oxidising agent, therefore it is unlikely that the reaction may occur between PbO_2 and HNO_3.

However, the acid-base reaction occurs between PbO and HNO_3 as

$$2\,PbO + 4\,HNO_3 \longrightarrow 2Pb(NO_3)_2 + 2H_2O$$

It is the passive nature of PbO_2 against HNO_3 that makes the reaction different from the one that follows with HCl.

16. (i) In this redox reaction, H in $LiAlH_4$ gets oxidised because of the addition of oxygen atom that leads to the formation of OH^-. Propanone (CH_3COCH_3) gets reduced because of addition of hydrogen atom to give propan-2-ol ($CH_3CH(OH)CH_3$).

(ii) This is not a redox reaction as neither hydrogen or oxygen or electron is removed or added.

(iii) This is not a redox reaction as neither hydrogen or oxygen or electron is removed or added.

Topic Practice 2

1. $Fe \longrightarrow Fe^{2+} + 2e^-$ (Oxidation)

$\frac{1}{2}O_2 + H_2O + 2e^- \longrightarrow 2OH^-$ (Reduction)

2. In MnO_4^{2-}, the oxidation number of Mn is +6. It can increase its oxidation number (to + 7) or decrease its oxidation number (to + 4, + 3, + 2, 0). Hence, it undergoes disproportionation reaction in acidic medium.

$$\overset{+6}{3MnO_4^{2-}} + 4H^+ \longrightarrow \overset{+7}{3MnO_4^-} + \overset{+4}{MnO_2} + 2H_2O$$

(On increases by 1 per atom (oxidation); On decreases by 2 per atom (reduction))

In MnO_4^-, Mn is in its highest oxidation state, i.e. +7. It can only decrease its oxidation number. Hence, it cannot undergo disproportionation reaction.

3. Cu^+ undergoes disproportionation to form Cu^{2+} and Cu.

$$2Cu^+(aq) \longrightarrow Cu^{2+}(aq) + Cu(s)$$

Thus, Cu^+ or Cu_2O acts both as an oxidant as well as reductant.

(i) When heated in air, Cu_2O is oxidised to CuO.

$$\overset{+1}{Cu_2}\overset{-2}{O} + \frac{1}{2}\overset{0}{O_2} \longrightarrow 2\overset{+2}{Cu}\overset{-2}{O}$$

i.e. Cu_2O acts as a reductant and reduces O_2 to O^{2-}.

(ii) When heated with Cu_2S, it oxidises S^{2-} to SO_2 and hence, Cu_2O acts as an oxidant.

$$2\overset{+1}{Cu_2}\overset{-2}{O} + \overset{+1}{Cu_2}\overset{-2}{S} \longrightarrow 6\overset{0}{Cu} + \overset{+4}{SO_2}$$

4. $\overset{0}{Cl_2}(g) + 2\overset{-2}{O}\overset{+1}{H}^-(aq) \longrightarrow \overset{+1}{Cl}\overset{-2}{O}^-(aq) + \overset{-1}{Cl}^-(aq) + \overset{+1}{H_2}\overset{-2}{O}(l)$

In this reaction, oxidation number of Cl increases from 0 (in Cl_2) to 1 (in ClO^-) as well as decreases from 0 (in Cl_2) to −1 (in Cl^-). So, it acts as both reducing as well as oxidising agent. This is an example of disproportionation reaction. In this reaction, ClO^- species bleaches the substances due to its oxidising action. [In hypochlorite ion (ClO^-), Cl can decrease its oxidation number from +1 to 0 or −1.]

Note Disproportionation reactions are a special type of redox reactions in which an element in one oxidation state is simultaneously oxidised and reduced.

5. (i) Combination reaction
(ii) Displacement reaction
(iii) Decomposition reaction
(iv) Metal displacement reaction
(v) Non-metal displacement reaction
(vi) Disproportionation reaction

6. $MnO_4^- + 8H^+ + ne^- \longrightarrow Mn^{2+} + 4H_2O$

$-1 + 8 + n = +2$

$-1 - 2 + 8 + n = 0$

$n = -5$ or $5e$

7. (i) $P_4(s) + 3OH^-(aq) + 3H_2O \longrightarrow PH_3(g) + 3H_2PO_2^-(aq)$

(ii) $3N_2H_4(l) + 4ClO_3^-(aq) \longrightarrow 6NO(g) + 4Cl^-(aq) + 6H_2O(l)$

(iii) $Cl_2O_7(g) + 4H_2O_2(aq) + 2OH^-(aq) \longrightarrow 2ClO_2^-(aq) + 5H_2O(l) + 4O_2(g)$

8. The skeleton equation for the given reaction is

$$MnO_4^-(aq) + SO_2(g) \longrightarrow Mn^{2+}(aq) + HSO_4^-(aq)$$

On solving similar to example 12 of page 307, we get following balanced equation.

$$2MnO_4^-(aq) + 5SO_2(g) + 2H_2O(l) + H^+(aq) \longrightarrow 2Mn^{2+}(aq) + 5HSO_4^-(aq)$$

This represents the final balanced redox equation.

9. If no salt-bridge is used, the positive ions (i.e. Zn^{2+}) formed by loss of electrons will accumulate around the zinc electrode and negative ions (i.e. SO_4^{2-}) left after reduction of Cu^{2+} ions will accumulate around the copper electrode. Thus, the solution will develop charges. Further, since the inner circuit is not complete, the current stops flowing.

10. KCl cannot be used as an electrolyte in the salt-bridge because Cl^- ions will combine with Ag^+ ions to form white precipitate of AgCl.

11. (i) At anode, $Zn \longrightarrow Zn^{2+} + 2e^-$

At cathode, $Br_2 + 2e^- \longrightarrow 2Br^-$

$$Zn + Br_2 \longrightarrow Zn^{2+} + 2Br^-$$

(ii) At anode, $[Cr \longrightarrow Cr^{3+} + 3e^-] \times 2$

At cathode, $[I_2 + 2e^- \longrightarrow 2I^-] \times 3$

$$2Cr + 3I_2 \longrightarrow 6I^- + 2Cr^{3+}$$

(iii) At anode, $H_2(g) \longrightarrow 2H^+(aq) + 2e^-$

At cathode, $Cu^{2+}(aq) + 2e^- \longrightarrow Cu(s)$

$$H_2(g) + Cu^{2+}(aq) \longrightarrow Cu(s) + 2H^+(aq)$$

Cathode will be positive terminal in each cell.

12. $E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode}$

$= E^\circ_{Ag^+/Ag} - E^\circ_{Cu^{2+}/Cu} = +0.80 - (+0.34) = 0.46\ V$

13. Since, the reduction potential of copper is less than that of Ag, so Cu electrode behaves as anode and Ag electrode as cathode.

$$Cu \mid Cu^{2+}(aq) \parallel Ag^+(aq) \mid Ag$$

14. Since, reduction potential of silver is more than that of hydrogen ($E^\circ_{H^+|H_2}$, Pt = 0), silver vessel will be suitable to store 1M HCl.

On the other hand, $E^\circ_{Al^{3+}|Al}$ is less than that of hydrogen $E^\circ_{H^+|H_2}$, so hydrogen will get liberated if stored in aluminium vessel.

15. $E^\circ_{(Fe^{3+}|Fe^{2+})}$ is lower than that of $E^\circ_{(Br|Br^-)}$. Therefore, Fe^{2+} can reduce Br_2 but Br^- cannot reduce Fe^{3+}. Thus, Fe^{3+} cannot oxidise Br^- to Br_2.

16. If H^+ or any acid appears on either side of the chemical equation, the reaction takes place in the acidic solution. If OH^-, or any base, appears on either side of the chemical equation, the solution is basic. If neither H^+, OH^- nor any acid or base is present in the chemical equation, the solution is neutral.

$E^\circ_{Al^{3+}/Al} = -1.66$ V, $E^\circ_{Cu^{2+}/Cu} = +0.34$ V,

$E^\circ_{Zn^{2+}/Zn} = -0.44$ V, $E^\circ_{Mg^{2+}/Mg} = -2.36$ V

and $E^\circ_{Fe^{2+}/Fe} = -0.76$ V

A metal with more negative value of E°_{red} is a stronger reducing agent than those which have less negative or positive value of E°_{red}.

Therefore, Mg can displace all the given metals from their aqueous salt solutions. Al can displace all metals except Mg from their aqueous salt solutions. Zinc can displace Fe and Cu from their aqueous salt solutions and Fe can displace only Cu from its aqueous salt solution.

Hence, the order in which they can displace each other from the solution of their salts is as follows

Mg, Al, Zn, Fe, Cu

18. Since $Mg^{2+}(aq)$ | Mg electrode = −2.36 V is at lower potential than $Al^{3+}(aq)$ | Al electrode = − 1.66 V, therefore, $Mg^{2+}(aq)$ | Mg electrode acts as the anode and $Al^{3+}(aq)$ | Al acts as the cathode.

In other words, Mg loses electrons and Al^{3+} ion accepts electrons. Thus, the cell reaction is

$$3\,Mg + 2\,Al^{3+} \longrightarrow 3\,Mg^{2+} + 2\,Al$$

and $E^\circ_{cell} = E^\circ_{Al^{3+}|Al} - E^\circ_{Mg^{2+}|Mg}$

$= -1.66 - (-2.36) = +0.70$ V

19. $(CN)_2(g) + 2OH^-(aq) \longrightarrow CN^-(aq) + CNO^-(aq) + H_2O(l)$

(i) Let the oxidation number of C in $(CN)_2$ be x.

$2x + 2(-3) = 0$ or $x = +3$

(ii) Let the oxidation number of C in CN^- be x.

$x + (-3) = -1$ or $x = +2$

(iii) Let the oxidation number of C in CNO^- be x.

$x + (-3) + (-2) = -1$ or $x = +4$

$$\overset{-3}{(CN)_2}(g) + 2OH^-(aq) \longrightarrow \overset{+2}{CN^-}(aq) + \overset{+4}{CNO^-}(aq) + H_2O(l)$$

Oxidation no. decreases

Oxidation no. increases

The following information we can drawn from the above reaction.

(i) Decomposition of cyanogen in the cyanide ion (CN^-) and cyanate ion (CNO^-) occurs in basic medium.

(ii) Cyanogen $(CN)_2$ acts as both reducing agent as well as oxidising agent.

(iii) The reaction is an example of disproportionation reaction (a special type of redox reaction).

(iv) Cyanogen $(CN)_2$ is called pseudohalogen while CN^-, CNO^- ions are called pseudohalide ions.

20.
$$\overset{0}{Cl_2}(aq) + \overset{+4}{SO_2}(aq) + H_2O(l) \longrightarrow \overset{-1}{Cl^-}(aq) + \overset{+6\ 2-}{SO_4}(aq)$$

ON decreases by 1 per Cl-atom

ON increases by 2 per S-atom

Multiply Cl^- by 2 because in Cl_2 there are two chlorine atoms.

$$Cl_2(aq) + SO_2(aq) + H_2O(l) \longrightarrow 2\,Cl^-(aq) + SO_4^{2-}(aq)$$

Balance first charge by adding $4H^+$ to RHS and then multiply H_2O by 2.

$$Cl_2(aq) + SO_2(aq) + 2H_2O(l) \longrightarrow 2Cl^-(aq) + SO_4^{2-} + 4H^+$$

This represents the balanced redox reaction.

21. Number of millimoles of $K_2Cr_2O_7$ present in 24 mL of 0.5 M Mohr's salt solution $= 24 \times 0.5 = 12$

The balanced chemical equation for the redox reaction is

$$K_2Cr_2O_7 + 6\,(NH_4)_2SO_4 \cdot FeSO_4 \cdot 6H_2O + 7H_2SO_4 \longrightarrow K_2SO_4 + 6\,(NH_4)_2SO_4 + 3\,Fe_2(SO_4)_3 + Cr_2(SO_4)_3 + 43\,H_2O$$

From the balanced equation, 6 moles Mohr's salt are oxidised by 1 mole of $K_2Cr_2O_7$.

∴ 12 millimoles of Mohr's salt will be oxidised by

$$= \frac{1}{6} \times 12 = 2 \text{ millimoles } K_2Cr_2O_7$$

22. $4NH_3(g) + 5O_2(g) \xrightarrow[Pt]{1100K} 4NO(g) + 6H_2O(g)$

$4 \times 17 = 68$ g; $5 \times 32 = 160$ g; $4 \times 30 = 120$ g

68 g NH_3 reacts with = 160 g O_2

1 g NH_3 reacts with $= \frac{160 \times 1}{68}$ g O_2

∴ 10 g NH_3 reacts with $= \frac{160 \times 10}{68} = 23.5$ g O_2

But available amount of O_2 is 20.0 g which is less than the amount which is required to react with 10 g NH_3. So, O_2 is the limiting reagent and it limits the amount of NO produced. From the above balanced equation,

160 g of O_2 produces 120 g NO

1 g of O_2 produces $\frac{120 \times 1}{160}$ g NO

∴ 20 g of O_2 will produce $\frac{120 \times 1 \times 20}{160} = 15$ g NO

23.

Element	F	Cl	Br	I
Standard reduction potential ($E°$/V)	+ 2.87	+ 1.36	+ 1.09	+ 0.54

Standard reduction potentials of halogens are positive and decrease from fluorine to iodine.

Thus, halogens act as strong oxidising agents and their oxidising power decreases from fluorine to iodine. Fluorine is the strongest oxidising agent, it oxidises other halide ions to halogens in solution or when they are dry.

$$F_2 + 2X^- \longrightarrow 2F^- + X_2 \quad (X^- = Cl^-, Br^- \text{ or } I^-)$$

In general, a halogen of low atomic number oxidises the halide ion of higher atomic number.

Halide ion, X^- or HX molecule when reacts with an oxidising agent, it reduces the oxidising agent and is itself oxidised to X_2 molecule. The tendency of X^- ions to lose electrons to form X_2 molecules increases from F^- to I^- ions. Due to this, reducing power of X^- ions or HX molecules increases from F^- to I^- to HI molecules.

Halide ions	I^-	Br^-	Cl^-	F^-
Electrode potential (V)	−0.54	−1.09	−1.36	−2.87

$$\underset{\text{Strongest reducing agent}}{HI} > HBr > HCl > \underset{\text{Weakest reducing agent}}{HF}$$

(reducing power decreases)

$$2HF + H_2SO_4 \longrightarrow \text{No reaction}$$

(HCl also does not react with H_2SO_4)

$$2HI + H_2SO_4 \longrightarrow SO_2 + 2H_2O + I_2$$

(HBr also reacts in the same manner)

HF or F^- is a weak reducing agent. It is so weak as a reducing agent that it does not reduce even very strong oxidising agent such as H_2SO_4 while HI or I^- reduces strong oxidising agent and is itself oxidised to I_2.

That's why, HI is the best reductant.

24. Since, $E°$ of Cu^{2+}/Cu electrode (+ 0.34 V) is higher than that of H^+/H_2 electrode (0.0 V), therefore, H^+ ions cannot oxidise Cu to Cu^{2+} ions and hence, Cu does not dissolve in dil. HCl.

In contrast, the electrode potential of NO_3^- ion, i.e. NO_3^-/NO electrode (+ 0.97 V) is higher than that of copper electrode and hence, it can oxidise Cu to Cu^{2+} ions and hence Cu dissolves in dil. HNO_3 due to oxidation of Cu by NO_3^- ions and not by H^+ ions.

Exam Pattern Solutions

Part-A

1. (*b*) Among the given species Li is the strongest reducing agent as its standard electrode reduction potential value is very negative.

2. (*b*) In reaction, $H_2O + \overset{0}{Br_2} \longrightarrow \overset{+1}{HO}\overset{-1}{Br} + H\overset{-1}{Br}$

The oxidation number of bromine increases from 0 (Br_2) to +1 (HOBr) and decreases from, 0 (Br) to −1 (HBr), hence bromine is both oxidised as well as reduced in given reaction.

3. (*b*) $\overset{+1\ +5\ -2}{KClO_3} \longrightarrow \overset{+1\ -1}{2KCl} + 3\overset{0}{O_2}$

The oxidation state of Cl decreases from +5 ($KClO_3$) to −1 (KCl), while that of oxygen increases from −2 ($KClO_3$) to 0 (O_2).

Thus, chlorine is reduced and oxygen is oxidised.

4. (*d*) The largest oxidation state is exhibited by $3d^5 4s^2$.

5. (*a*) $2H_2 + O_2 \longrightarrow 2H_2O$

In the above reaction electrons are transferred from H to O-atom.

6. (*d*) The oxidation number Cl is +5 in ClO_3^-.

$$x + 3(-2) = -1$$

$$\Rightarrow \quad x = +5$$

7. (*c*) In $S_4O_6^{2-}$, let 'x' be the oxidation number of S.

$$\therefore \quad 4x - 12 = -2$$

$$x = \frac{10}{4} = \frac{5}{2}$$

8. (*c*)

$$\overset{+7}{MnO_4^-} + 8H^+ + 5Br^- \longrightarrow Mn^{2+} + 4H_2O + \frac{5}{2}\overset{0}{Br_2}$$

(Oxidation above Br; Reduction below Mn)

Since, in this reaction Br^- reduces MnO_4^- into Mn^{2+} and itself gets oxidised into Br_2, so here Br^- act as the reducing agent.

9. (*b*) In reaction, $2H_2O(l) \xrightarrow{\Delta} 2H_2(g) + O_2(g)$, H_2O on heating will split into H_2 and O_2. Thus, it is an example of decomposition reaction.

10. (*d*) $\overset{+4}{2NO_2} + 2\overset{-}{O}H \longrightarrow \overset{+3}{NO_2^-} + \overset{+5}{NO_3^-} + H_2O$

This is an example of disproportionation reaction as oxidation state of N changes from +4 to +3 and +5, respectively.

11. (*d*) $8\overset{0}{Al} + 3\overset{8/3}{Fe_3O_4} \longrightarrow 4\overset{+3}{Al_2}\overset{0}{O_3} + 9\overset{0}{Fe}$

Oxidation half reaction.

$[2Al + 3H_2O \longrightarrow Al_2O_3 + 6e^- + 6H^-] \times 4$

Reduction half reaction.

$[3Fe_3O_4 + 8e^- + 8H^+ \longrightarrow 3Fe + 4H_2O] \times 3$

So, 24 electrons transferred.

12. (*c*) $Cr_2O_7^{2-} + 14H^+ + nFe^{2+} \longrightarrow 2Cr^{3+} + nFe^{3+} + 7H_2O$

$$Cr_2O_7^{2-} + 14H^+ + 6e^- \longrightarrow 2Cr^{3+} + 7H_2O \qquad \text{...(i)}$$

$$Fe^{2+} \longrightarrow Fe^{3+} + e^-$$

Multiply eq. (ii) by 6.

$$6Fe^{2+} \longrightarrow 6Fe^{3+} + 6e^-$$

Thus, balanced equation is

$$Cr_2O_7^{2-} + 14H^+ + 6Fe^{2+} \longrightarrow 2Cr^{3+} + 6Fe^{3+} + 7H_2O$$

$\therefore$ The value of n is 6.

13. (*a*) Salt-bridge is a U-tube containing a solution of potassium chloride or ammonium nitrate usually solidified by boiling with *agar-agar*.

14. (*d*) Among the given cases, reaction does not occurs between Ag and Fe^{3+}. This is because $E°$ value of this reaction is comes out to be − 0.003 V.

15. (*b*) The negative $E°$ value indicates that redox couple is stronger reducing agent than H^+/H_2 couple.

16. (*c*) More negative is the $E°$ value greater is the reducing power. Thus, decreasing order of reducing power is

I > IV > II > III.

17. (*a*) Both (A) and (R) are correct and (R) is the correct explanation of (A).

18. (*a*) Both (A) and (R) are correct and (R) is the correct explanation of (A).

19. (*a*) Both (A) and (R) are correct and (R) is the correct explanation of (A).

20. (*b*) Fluorine has the maximum $E°$ value among the halogens, hence it reduces first than other halogen and acts as best oxidant. Fluorine is most electronegative atom due to its small size. Thus, both statement are correct but (R) is not the correct explanation of (A).

21. (c) (A) is correct while (R) is incorrect. Correct (R) is Fe is a better reducing agent than H_2.

22. (c) (A) is correct while (R) is incorrect. When Zn rod is dipped in $Cu(NO_3)_2$ solution, Zn gets oxidised and Cu^{2+} ions gets reduced.

$$\overset{+2}{Zn} + \overset{+2}{Cu}(NO_3)_2 \longrightarrow \overset{+2}{Zn}(NO_3)_2 + \overset{0}{Cu}$$

Reduced

Oxidised

23. (i) (d) Let the oxidation state of K in KO_2 be x.

$$\therefore \quad +1 + 2x = 0$$
$$2x = -1$$
$$x = -1/2$$

Thus, oxidation state of K in KO_2 is $-1/2$.

(ii) (a) Among the given elements, fluorine never has positive oxidation number in any of its compound.

(iii) (a) $[Fe(H_2O)_5NO]SO_4$

$$x + (+1) + 5(0) = +2$$
$$x = +1$$

Therefore, iron is present in +1 oxidation state.

Or (a) $2\overset{+7}{K}MnO_4 + 5H_2C_2O_4 + 3H_2SO_4 \longrightarrow K_2SO_4 + 2\overset{+2}{Mn}SO_4 + 10CO_2 + 8H_2O$

Thus, when $KMnO_4$ is reduced with oxalic acid in acidic medium, the oxidation number of Mn changes from +7 to +2.

(iv) (c) Let, the oxidation state of sulphur in $Na_2S_4O_6$ be x.

$$(+1) \times 2 + 4(x) + (-2) \times 6 = 0$$
$$2 + 4x - 12 = 0$$
$$x = \frac{10}{4} = 2.5$$

24. (i) (a) Both (A) and (R) are correct and (R) is the correct explanation of (A).

(ii) (a) Both (A) and (R) are correct and (R) is the correct explanation of (A).

(iii) (c) (A) is correct and (R) is incorrect.

The Br_3O_8 compound has two types of oxidation states, i.e. +6 and +4. Average of three oxidation number of Br is $\frac{6+6+4}{3} = \frac{16}{3}$

$$x + \frac{16}{3} = 0 \Rightarrow x = \frac{-16}{3}$$

(iv) (c) $2\overset{+7}{MnO_4^-} + H_2O + I^- \longrightarrow 2\overset{+7}{MnO_2} + 2\overset{-}{OH} + IO_3^-$

By considering the above reaction it is clear that the (A) is correct and (R) is incorrect. The correct (R) is given as,

The oxidation state of Mn in given reaction is changes from +7 to +4.

Or (c) (A) is correct and (R) is incorrect. A metal with lower SRP (Standard reduction potential) can have the ability to displace metal with higher SRP from their solutions.

As $E^\circ_{Zn^{2+}/Zn} = -0.76\,V$; $E_{Fe^{2+}/Fe} = -0.44\,V$

Part-B

1. (i) Suppose that the oxidation number of P in HPO_3^{2-} be x.

Then, $1 + x + 3(-2) = -2$ or $x + 1 - 6 = -2$

or $x = +3$

(ii) Suppose that the oxidation number of P in PO_4^{3-} be x.

Then, $x + 4(-2) = -3$ or $x - 8 = -3$

or $x = +5$

2. In a disproportionation reaction, the same species simultaneously get oxidised as well as reduced. Therefore, for such a redox reaction to occur, the reacting species must contain an element which has atleast three oxidation states.

The element, in reacting species, is present in an intermediate state while lower and higher oxidation states are available for reduction and oxidation to occur (respectively).

Fluorine is the strongest oxidising agent. It does not show positive oxidation state. That's why fluorine does not show disproportionation reaction.

3. $\overset{+1\ -2}{H_2O}(s) + \overset{0}{F_2}(g) \longrightarrow \overset{+1\ -1}{HF}(g) + \overset{+1\ -2\ +1}{HOF}(g)$

In the above reaction, as the oxidation state of F decreases from 0 to −1 and increases from 0 to +1. That means, F is both oxidised and reduced. Hence, the given reaction is a redox reaction.

4. PbO is a base. It reacts with nitric acid and forms soluble lead nitrate.

$$PbO + 2HNO_3 \longrightarrow Pb(NO_3)_2 + H_2O$$

(Acid - base reaction) Soluble

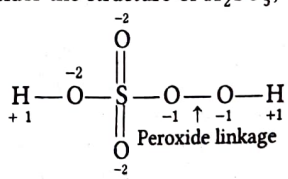
Nitric acid does not reacts with PbO_2. Both of them are strong oxidising agents. In HNO_3, nitrogen is in its maximum oxidation state (+5) and in PbO_2, lead is in its maximum oxidation state (+4). Therefore, no reaction takes place.

5. (i) Oxidation number of S in H_2SO_5 is

$$2(+1) + x + 5(-2) = 0 \text{ or } x = +8$$

But oxidation number of S cannot be more than 6 because S has only 6 valence electrons. This misconception is removed by calculating oxidation number of S by chemical bonding method.

Let us consider the structure of H_2SO_5,



In H_2SO_5, two oxygen atoms are in −1 oxidation state and other three are in −2 oxidation state

Let the oxidation number of S be x.

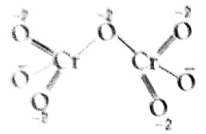
$$\underset{\text{For H}}{2(+1)} + \underset{\text{For three O}}{3(-2)} + x + \underset{\text{For O — O}}{2(-1)} = 0$$
$$2 + (-6) + x + (-2) = 0$$
$$x = +6$$

Therefore, the oxidation number of S in H_2SO_5 is + 6.

(ii) Oxidation number of Cr in $Cr_2O_7^{2-}$ is

$$2x + (-2)7 = -2$$
$$2x = +12$$
$$x = +6$$

Let us consider the structure of $Cr_2O_7^{2-}$ ion



Let the oxidation number of each Cr-atom be x.

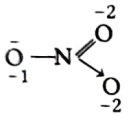
$$4(-2) + (-2) + 2(-1) + 2x = 0$$
$$-8 - 2 - 2 + 2x = 0$$
$$2x = +12,\ x = +6$$

Oxidation number of Cr in $Cr_2O_7^{2-}$ is same, i.e. + 6 whether it is calculated by conventional method or by chemical bonding method. Hence, there is no fallacy.

(iii) Oxidation number of N in NO_3^- be x.

$[x + 3(-2) = -1$ or $x = +5]$

Let us consider the structure of NO_3^- ion.



Let the oxidation number of N be x.

$$1(-1) + x + 1(-2) + 1(-2) = 0$$
For $\bar{O}$ For (= O) For ($\rightarrow$ O)

$$x = +5$$

Oxidation number of N in NO_2^- ion is same, i.e. + 5 whether it is calculated by conventional method or by chemical bonding method.

Hence, there is no fallacy.

6. (i) Hg (II) Cl_2 (ii) Ni (II) SO_4
(iii) Sn (IV) O_2 (iv) Th_2(I) SO_4
(v) Fe_2(III)$(SO_4)_3$ (vi) Cr_2(III) O_3

7. (i) In SO_2, S is in + 4 oxidation state. It can have minimum oxidation number − 2 and maximum oxidation number is + 6.

Therefore, S in SO_2 can either decrease or increase its oxidation number. So, SO_2 can act both as an oxidising as well as reducing agent.

(ii) In H_2O_2, O is in −1 oxidation state. It can have minimum oxidation number − 2 and maximum oxidation number zero.

Therefore, O in H_2O_2 can either decrease or increase its oxidation number. So, H_2O_2 can act both as an oxidising as well as reducing agent.

(iii) In O_3 ,O is in zero oxidation state. It cannot increase its oxidation number, it can only decrease its oxidation number from zero to −1 or −2. So, ozone can act only as an oxidising agent.

(iv) In HNO_3, the oxidation number of N is + 5. It is maximum. So, N in HNO_3 can only decrease its oxidation number. So, it can act as an oxidising agent only.

8. (i) + 9 charge on the left, +1 charge on the right; add 8 electrons to the left side.

(ii) 0 charge on the left, + 6 charge on the right; add 6 electrons on the right side.

9. The skeletal equation is,

$$Mn^{3+}(aq) \longrightarrow Mn^{2+}(aq) + MnO_2(s) + H^+(aq)$$

After balancing above equation we get,

$$2Mn^{3+}(aq) + 2H_2O(l) \longrightarrow Mn^{2+} + MnO_2(s) + 4H^+(aq)$$

10. Since, $E°$ of Fe^{2+} /Fe (− 0.44 V) is lower than that of Ag^+ / Ag (+ 0.80 V) electrode, therefore, Ag^+ gets reduced and Fe gets oxidised. As a result, concentration of Ag^+ ions decreases while that of NO_3^- ions remain unchanged.

$$2Ag^+(aq) + Fe(s) \longrightarrow 2Ag(s) + Fe^{2+}(aq)$$

11. (i) Since, $E°$ of Zn is more negative than that of Fe, therefore, Zn will be oxidised to Zn^{2+} ions while Fe^{2+} ions will be reduced to Fe. In other words, Fe will not reduce Zn^{2+} ions.

(ii) Since, $E°$ of Fe is more negative than that of Ni, therefore, Fe will be oxidised to Fe^{2+} ions while Ni^{2+} ions will be reduced to Ni. Thus, Fe reduces Ni^{2+} ions.

12. Measure the electrode potential of the given species by connecting the redox couple of the given species with standard hydrogen electrode.

If it is positive, the electrode of the given species acts as reductant and if it is negative, it acts as an oxidant.

Find the electrode potentials of the other given species in the same way, compare the values and determine their comparative strength or an reductant or oxidant. e.g. Measurement of standard electrode potential of Zn^{2+}/Zn electrode using SHE as a reference electrode.

13. The given redox reaction for the galvanic cell is

$$Zn(s) + 2Ag^+(aq) \longrightarrow Zn^{2+}(aq) + 2Ag(s)$$

At anode, Zn is oxidised to Zn^{2+} ions and at cathode, Ag^+ ions are reduced to Ag metal.

Thus, galvanic cell for the above redox reaction may be depicted as $Zn \mid Zn^{2+}(aq) \parallel Ag^+(aq) \mid Ag$

(i) Zn electrode is negatively charged because of the oxidation of Zn to Zn^{2+} ions, electrons are accumulated on zinc electrode.

(ii) The ions carry current in the cell. Current flows from Ag electrode to Zn electrode. While electrons flow from Zn electrode to Ag electrode.

(iii) Individual reaction at each electrode

At anode, $Zn \longrightarrow Zn^{2+} + 2e^-$

At cathode, $Ag^+(aq) + e^- \longrightarrow Ag(s)$

14. Refer to text on pages 294 and 295.

(i) Carbon has a high affinity towards oxygen so it is generally used as reducing agent. However, in case of highly reactive metals, it cannot be used as a reducing agent.

(ii) Ozone being a source of oxygen generally behaves as an oxidising agent.

(iii) As, we know addition of hydrogen is called reduction. So, nascent hydrogen behaves as a reducing agent.

(iv) Nitric acid is a good oxidant, i.e. oxidising agent as it readily provides oxygen.

(v) Chlorine can behave as oxidant as well as reductant depending upon the nature of other reactant.

5. **Oxidants** (is) H^+ (ii) H_2O_2 (iii) $[Fe(CN)_6]^{3-}$ (iv) F_2 (v) $NaClO_3$

Reductants (i) Zn (ii) $[Fe(CN)_6]^{4-}$ (iii) H_2O_2 (iv) BrO_3^- (v) I_2

(For details see electron transfer concept given on page 296).

6. (i) **KI_3**

$\overset{+1}{K}\overset{x}{I_3}$

$1(+1) + 3x = 0$ or $x = -1/3$

Therefore, the average oxidation number of I is $-\frac{1}{3}$. It is wrong. Let us consider the structure of KI_3.

$\overset{+1}{K}\,(I - I \leftarrow I)^{-1}$, in this structure, coordinate bond is formed between I_2 molecule and I^- ion. Hence, the oxidation number of three I-atoms in KI_3 are 0,0 (in I_2) and -1 respectively.

(ii) **$H_2S_4O_6$**

Let the oxidation number of S be x.

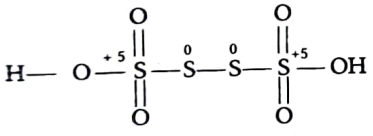
$\overset{+1}{H_2}\overset{x}{S_4}\overset{-2}{O_6}$

$2(+1) + 4x + 6(-2) = 0$

$4x = +10$ or $x = +\frac{10}{4} = +2.5$

Therefore, the average oxidation state of S be 2.5

Let us consider the structure of $H_2S_4O_6$.



In $H_2S_4O_6$, the oxidation number of each of two S-atoms which are linked with each other by a single bond (in the centre) is zero and each of remaining two S-atoms is + 5. Hence, the oxidation number of 4 S-atoms in $H_2S_4O_6$ from left to right are + 5, 0, 0 and + 5 respectively.

(iii) **Fe_3O_4** Refer to text on page 300.

(iv) **$\underline{C}H_3\underline{C}H_2OH$**

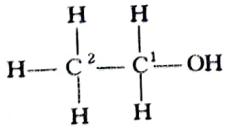
$\overset{x}{C_2}\overset{+1}{H_6}\overset{-2}{O}$

$2x + 6(+1) + 1(-2) = 0$

$2x = -4$ or $x = -2$

Therefore, the average oxidation number of C is -2.

Let us consider the structure of CH_3CH_2OH.



Oxidation number of C_1-atom

$= 1(+1) + 2(+1) + x + 1(-1) = 0$

[C_1-atom is attached to one CH_3 group (oxidation number = + 1), two H-atoms and one — OH group (oxidation number = -1)]

$x = -2$

Oxidation number of C_2-atom $= 3(+1) + x + 1(-1) = 0$

[C_2-atom is attached to three H-atoms and one — CH_2OH group (oxidation number = -1)]

$x = -2$

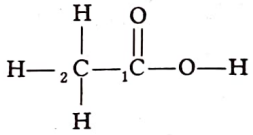
(v) **$\underline{C}H_3\underline{C}OOH$**

$\overset{x}{C_2}\overset{+1}{H_4}\overset{-2}{O_2}$

$2x + 4(+1) + 2(-2) = 0$

$2x + 4 + (-4) = 0$ or $x = 0$

Therefore, the average oxidation number of C is zero. Let us consider the structure of CH_3COOH.



Oxidation number of C_1-atom

$= 1(+1) + x + 1(-2) + 1(-1) = 0$

[C_1-atom is attached to one —CH_3 group (oxidation number = + 1), one oxygen atom and one — OH group (oxidation number = -1)]

$x = +2$

Similarly, oxidation number of C_2-atom

$3(+1) + x + 1(-1) = 0$

$\Rightarrow$ $x = -2$

17. (i) $CuO(s) + H_2(g) \longrightarrow Cu(s) + H_2O(g)$

Assign oxidation numbers of each atom above its symbol.

$\overset{+2\,-2}{CuO}(s) + \overset{0}{H_2}(g) \longrightarrow \overset{0}{Cu}(s) + \overset{+1\,-2}{H_2O}(g)$

Oxidation number of Cu in CuO is + 2. It decreases from + 2 to zero in Cu. While oxidation number of hydrogen increases from 0 (in H_2) to + 1 (in H_2O). This shows that CuO is reduced to Cu but H_2 is oxidised to H_2O. Hence, it is an example of redox reaction.

(ii) $\overset{+3\,-2}{Fe_2O_3}(s) + 3\overset{+2\,-2}{CO}(g) \longrightarrow 2\overset{0}{Fe}(s) + \overset{+4\,-2}{CO_2}(g)$

Oxidation number of Fe decreases from + 3 (in Fe_2O_3) to zero (in Fe) and oxidation number of C increases from + 2 (in CO) to + 4 (in CO_2). This shows that Fe_2O_3 is reduced to Fe and CO is oxidised to CO_2. Hence, it is a redox reaction.

(iii) $4\overset{+3\,-1}{BCl_3}(g) + 3\overset{+1\,+3\,-1}{LiAlH_4}(s) \longrightarrow$

$\overset{-3\,+1}{2B_2H_6}(g) + 3\overset{+1\,-1}{LiCl}(s) + 3\overset{+3\,-1}{AlCl_3}(s)$

Oxidation number of B decreases from + 3 (in BCl_3) to -3 (in B_2H_6) and oxidation number of H increases from -1 (in $LiAlH_4$) to + 1 (in B_2H_6). This shows that BCl_3 is reduced to B_2H_6 and $LiAlH_4$ is oxidised. Hence, it is redox reaction.

(iv) $\overset{0}{2K(s)} + \overset{0}{F_2} \longrightarrow \overset{+1\,-1}{2KF(s)}$

Oxidation number of K increases from zero (in K) to + 1 (in KF) and oxidation number of F reduces from zero (in F_2) to – 1 (in KF). This shows that K is oxidised and F_2 is reduced. Hence, it is a redox reaction.

(v) $\overset{-3\,-1}{4NH_3(g)} + 5\overset{0}{O_2(g)} \longrightarrow \overset{+2\,-2}{4NO(g)} + 6\overset{+1\,-2}{H_2O(g)}$

Oxidation number of N increases from – 3 (in NH_3) to + 2 (in NO) and oxidation number of O decreases from zero (in O_2) to – 2 (in NO and H_2O). This shows that NH_3 is oxidised and O_2 is reduced. Hence, it is a redox reaction.

18. (i) Reduction

$$\overset{-2}{H_2S} + Fe^{3+} \longrightarrow Fe^{2+} + \overset{0}{S} + H^+$$

Oxidation

Here, H_2S is oxidised to sulphur (S), and Fe^{3+} is reduced to Fe^{2+}.

The balanced chemical equation is given as

$$2Fe^{3+} + H_2S \longrightarrow 2Fe^{2+} + S + 2H^+$$

(ii) Reduction

$$I^- + \overset{+5}{IO_3^-} + H^+ \longrightarrow \overset{0}{I_2} + H_2O$$

Oxidation

Here, I^- is oxidised to I_2 and IO_3^- is reduced to I_2.

On solving, we get the following balanced equations

$$2IO_3^- + 12I^- + 12H^+ \longrightarrow 7I_2 + 6H_2O$$

(iii) Reduction

$$\overset{0}{Bi(s)} + \overset{+5}{NO_3^-} + H^+ \longrightarrow \overset{+4}{NO_2} + Bi^{3+} + H_2O$$

Oxidation

Here, Bi(s) is oxidised to Bi^{3+}, while NO_3^- is reduced to NO_2.

The balanced chemical equation is given as

$$Bi(s) + 3NO_3^- + 6H^+ \longrightarrow 3NO_2 + Bi^{3+} + 3H_2O$$

(iv) Reduction

$$I^- + \overset{0}{O_2(g)} + H_2O \longrightarrow \overset{0}{I_2} + \overset{-2}{OH^-}$$

Oxidation

Here, I^- is oxidised to I_2 and $O_2(g)$ is reduced to OH^-.

$$4I^- + O_2(g) + 2H_2O \longrightarrow 4OH^- + 2I_2$$

19. (i) $\overset{+7}{MnO_4^-}(aq) + I^-(aq) \longrightarrow \overset{+4}{MnO_2}(s) + \overset{0}{I_2}(s)$

(in basic medium)

(a) Separate the equation into oxidation half reaction and reduction half reaction.

Oxidation half reaction,

$$I^-(aq) \longrightarrow I_2(s) \quad ...(i)$$

Reduction half reaction,

$$MnO_4^-(aq) \longrightarrow MnO_2(s) \quad ...(ii)$$

(b) To balance I-atoms in the oxidation half reaction, we write it as

$$2I^-(aq) \longrightarrow I_2(s) \quad ...(iii)$$

and to balance oxidation number add $2e^-$ to RHS

$$2I^-(aq) \longrightarrow I_2 + 2e^- \quad ...(iv)$$

There is no need to balance the charge (charge on either side of equation is balanced).

(c) To balance reduction half reaction, first balance oxidation number by writing required number of electrons to LHS and then balance charge by adding OH^- ions because reaction occurs in basic medium.

$$MnO_4^-(aq) + 3e^- \longrightarrow MnO_2(s) \quad ...(v)$$

$$MnO_4^-(aq) + 3e^- \longrightarrow MnO_2 + 4\,OH^- \quad ...(vi)$$

(d) Balance oxygen atoms by adding H_2O

$$MnO_4^- + 3e^- + 2H_2O \longrightarrow MnO_2 + 4\,OH^- \quad ...(vii)$$

By doing so, H atoms are automatically balanced.

(e) Now to equalise number of electrons we multiply oxidation half reaction (iv) by 3 and reduction half reaction (vii) by 2 respectively.

$$6I^-(aq) \longrightarrow 3\,I_2(s) + 6e^- \quad ...(viii)$$

$$2\,MnO_4^-(aq) + 6e^- + 4H_2O(l) \longrightarrow 2MnO_2(s) + 8OH^-(aq) \quad ...(ix)$$

(f) Add two half reactions to obtain net reactions after cancelling electrons on both sides.

$$2\,MnO_4^-(aq) + 6I^-(aq) + 4\,H_2O(l) \longrightarrow 2\,MnO_2(s) + 3\,I_2(s) + 8OH^-(aq)$$

This represents the final balanced redox reaction.

(ii) $MnO_4^-(aq) + SO_2(g) \longrightarrow Mn^{2+}(aq) + HSO_4^-$

(in acidic medium)

Reduction half reaction

$$MnO_4^-(aq) + 8H^+(aq) + 5e^- \longrightarrow Mn^{2+}(aq) + 4H_2O(l) \quad ...(i)$$

Oxidation half reaction

$$SO_2(g) + 2H_2O(l) \longrightarrow HSO_4^-(aq) + 3H^+(aq) + 2e^- \quad ...(ii)$$

(Balance the oxidation number as in (i). After balancing the oxidation number, charge is balanced on either side of equation by adding H^+ ions because the reaction occurs in acidic medium).

Multiply equation (i) by 2 and equation (ii) by 5 and add, we get

$$2MnO_4^-(aq) + H^+(aq) + 5SO_2(g) + 2H_2O(l) \longrightarrow 2Mn^{2+}(aq) + 5HSO_4^-(aq)$$

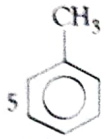
(iii) By following the steps discussed in (i), we get

$$H_2O_2(aq) + 2\,Fe^{2+}(aq) + 2H^+(aq) \longrightarrow 2\,H_2O(l) + 2\,Fe^{3+}(aq)$$

(iv) By following the steps discussed in (i), we get

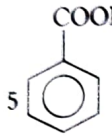
$$Cr_2O_7^{2-}(aq) + 2H^+(aq) + 3SO_2(g) \longrightarrow 2Cr^{3+}(aq) + 3SO_4^{2-}(aq) + H_2O(l)$$

20. (i) Oxidation of toluene to benzoic acid in acidic medium.



Toluene

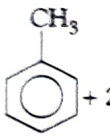
$$5 + 6MnO_4^-(aq) + 18H^+(aq) \longrightarrow$$



Benzoic acid

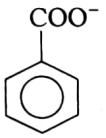
$$5 + 6Mn^{2+}(aq) + 14H_2O(l)$$

Oxidation of toluene to benzoic acid in basic and neutral medium.



Toluene

$$+ 2MnO_4^-(aq) \longrightarrow$$



Benzoate ion

$$+ 2MnO_2(s) + H_2O(l) + OH^-(aq)$$

On industrial scale, alcoholic potassium permanganate is preferred to acidic or alkaline potassium permanganate because in the presence of alcohol, both the reactants $KMnO_4$ and $C_6H_5CH_3$ are mixed very well and form homogeneous solution.

In homogeneous medium, reaction takes place faster than in heterogeneous medium. Further more in neutral medium, OH^- ions are produced in the reaction itself.

(ii) $\underset{\substack{\text{Sodium chloride}\\\text{(from inorganic mixture)}}}{2NaCl} + 2H_2SO_4 \longrightarrow 2NaHSO_4 + 2HCl$

HCl is a weak reducing agent. It cannot reduce H_2SO_4 to SO_2, that's why pungent smelling gas HCl is obtained.

$$\underset{\substack{\text{Sodium bromide}\\\text{(from inorganic mixture)}}}{2NaBr} + 2H_2SO_4 \longrightarrow 2NaHSO_4 + 2HBr$$

$$2HBr + H_2SO_4 \longrightarrow \underset{\text{Red vapours}}{Br_2\uparrow} + SO_2\uparrow + 2H_2O$$

HBr is a strong reducing agent, it reduces H_2SO_4 to SO_2 and is itself oxidised to Br_2. That's why we get red vapours of bromine when conc. H_2SO_4 reacts with inorganic mixture containing bromide salt.

21. (i) The electrochemical series or redox activity is based on the decreasing order of reduction potentials. This means that the species which gets reduced is higher in the electrochemical series as compared to the other which is to be oxidised (lose electrons).

In reaction (a), B^+ gets reduced by A and therefore B is higher than A in electrochemical series.

In reaction (b), D^+ gets reduced by B and therefore, D is higher in the electrochemical series than B.

In reaction (c), C^+ does not get reduced by D, therefore, C is lower than D in electrochemical series. But according to reaction (d), C^+ gets reduced by B and therefore, C is higher in electrochemical series than B. Thus, the correct order is $D > C > B > A$.

(ii) Both reactions do not occur because A cannot be reduced by C as well as D.

22. (i) The balanced chemical equation is

$$K_2Cr_2O_7 + 6KI + 12H_2SO_4 \longrightarrow K_2SO_4 + Cr_2(SO_4)_3 + H_2O + 3I_2 + 6KOH$$

Hence, 3 moles of iodine will be liberated.

(ii) In a redox reaction, the substance which oxidises the other or which itself undergo reduction is called the oxidising agent.

e.g. $Fe + Cu^{2+} \longrightarrow Fe^{2+} + Cu$

Here, Cu^{2+} gets reduced to Cu and oxidises Fe. So, Cu^{2+} is an oxidising agent.

(iii) In alkaline medium balanced equation is

$$2ClO_2 + 5H_2O_2 + 2OH^- \longrightarrow 2Cl^- + 5O_2 + 6H_2O$$

2 moles of ClO_2 oxidise 5 moles of H_2O_2

1 mole of ClO_2 will oxidise $= \frac{5}{2} \times 1 = 2.5$ mol

(iv) In oxidation → oxidation state increases.

In reduction → oxidation state decreases.

Or

HNO_2

$1 + x + 2(-2) = 0$

$x - 3 = 0$

$x = +3$

23.. (i) $KMnO_4$ is the compound of manganese which has highest oxidation state.

$KMnO_4(+1) + x + 4(-2) = 0$

$x = +7$

(ii) Fluorine is the most electronegative element. Hence, does not possess positive oxidation number in any of its compound.

(iii) The brown ring complex is $[Fe(H_2O)_5NO]SO_4$. In this oxidation number of Fe is +3.

(iv) $\overset{(+2)}{Mn}^{2+} + 2NO_3^- + 2OH^- \longrightarrow \overset{(+6)}{MnO_4^{2-}} + 2H_2O + 2NO$

The oxidation number of Mn changes from +2 to +6.

Or Let the oxidation state of I in IPO_4 be x.

$x + (-3) = 0$. [PO_4^{3-} ion has charge equal to −3.]

$x = +3$